

Choupo

The Open Source Chemical Process Simulator

Developer Guide

Choupo-dev

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Preface — the contract this code makes with you

If you have written CFD glue for a CFD solver, or stared too long at a black-box solver, you already know why this project exists. Every Newton step is auditable. Every dispatch is explicit. Every dependency is in the C++ standard library. The price of that discipline is that contributors have to play by a few rules. They are short, and they are non-negotiable.

This guide is opinionated on purpose. Read §1 carefully — those three principles override everything else — and treat §28 as the project's only blunt list of don'ts.

If you are about to write a class rather than read one, go straight to §20: five rules, each stated with the defect that paid for it, because a rule with a scar is remembered and a rule without one is negotiated. §19 says where the class may live, and §21 says how its contract is made enforceable.

Independence and provenance. Choupo is an independent project. Its file-based case organization (`controlDict`-style dictionaries, `constant/` and time directories such as `0/`) was inspired in part by the conventions popularised by OpenFOAM. Choupo is not affiliated with, endorsed by, or sponsored by OpenCFD Ltd or the OpenFOAM Foundation. The modules closest in spirit are independent implementations, not adaptations: the dictionary parser is Choupo's own component-aware tokenizer; model selection uses explicit factory registrations (deliberately *not* runtime-selection macros, §6); the `Rosenbrock23` integrator implements the published Shampine & Reichelt (SIAM J. Sci. Comput. 18, 1997) formulas; and the time-directory writer implements the layout convention from scratch. No OpenFOAM source code was adapted or reused.

1 Philosophy — the contract we promise

What you'll learn. The three invariants that every contribution is judged against, in plain language.

1. **Pedagogical clarity beats elegance.** The simulator must be readable by a chemical-engineering student in their third year. No template metaprogramming, no auto-magic, no clever short-circuits. Every Newton iteration and every K-value must be inspectable in both the source and the run-time output.
2. **Source transparency — no hidden registration.** Every factory base class has an *explicit* `registerBuiltins()` called from `main.cpp`. No `REGISTER_TYPE(MyClass)` macros, no static initialisers, no linker-dependent constructors. The student grep-searches `registerBuiltins` and *sees* every built-in type.
3. **Dependency purity — no copyleft or restrictive libraries linked into the binary.** C++ 17, the C++ standard library, and GNU make. Nothing else. No Boost, no Eigen, no Sundials, no CMake. Hand-rolled Newton, Gauss elimination, RK4, Wegstein, Michelsen.

Key idea. An "industrial-grade" contribution that violates any of these will be rejected even if it converges better and runs faster.

2 Repository tour

What you'll learn. What lives where. Useful as a map; you'll come back to it.

```
Choupo/
|-- Makefile                        top-level: release/debug switch
|-- make/
|   |-- compiler.mk                 CXX, CXXFLAGS, PLATFORM detection
|   |-- rules.mk                    pattern rules + dependency tracking
|   `-- wasm.mk                     Emscripten WASM build (gui/)
|-- build/<PLATFORM>[Debug]/        all.o/.d/binary (gitignored)
|-- choupoSolve                  symlink -> last built binary
|-- etc/bashrc                      sourceable shell env
|-- bin/                            runCase, runApplication, listCases, cleanCase
|-- data/
|   `-- standards/                   the FROZEN, committee-managed reference tree
|       |-- components/              <name>.dat --- component catalogue (arity-1)
|       |-- species/                 <name>.dat --- model species (ions, aq. gases)
|       |-- chemistry/               real equilibria: salts, speciation, minerals
|       |-- assets/                  physical kit, FLAT with kind = consumer
|       |                             (membranes, materials, adsorbents, resins)
|       |-- utilities/               <name>.dat --- plant-utility catalogue
|       |-- parameters/              every pair catalogue (arity-2):
|       |                             NRTL/ UNIQUAC/ Wilson/ Henry/ SRK/
|       |                             Pitzer/ eNRTL/ solution/ UNIFAC/
|       `-- mixtures/                predefined mixtures (air, ...)
|-- docs/                           THIS FILE + userGuide.{md,tex,pdf}
|-- src/
|   |-- core/                        Dictionary, SimulationResult, Constants, Types
|   |-- streams/                     ProcessStream
|   |-- thermo/                      Component, ThermoPackage, Database +
|   |                                 phase/, vaporPressure/, activityCoefficient/,
|   |                                 equationOfState/, heatCapacity/, membrane/
|   |-- materials/                   Material, MaterialRegistry
|   |-- solver/                      NewtonRaphson, NewtonND, Wegstein, StabilityTest
|   |-- unitOperations/              UnitOperation base + per-category folders
|   |-- outerDriver/                 OuterDriver base + Sweep/Grid/Pareto/Optim
|   |-- postProcessing/              PostProcessor base + Sizing/Costing passes
|   |-- propertyOps/                 PropertyOperation base (choupoProps)
|   |-- control/                     Controller base + PID/Schedule (choupoCtrl)
|   |-- dynamicDriver/               ONE home of the time-integrated flowsheet driver
|   |                                 (choupoCtrl + choupoSemiContinuous; 2026-09-20)
|   |-- reporting/                   BalanceMath, EnergyBalanceReport (the ledger)
|   |-- result/                      the pipeline's OUTPUT records (SimulationResult,
|   |                                 UnitProfile, ResultEmitter) -- see Sec. layering
|   |-- io/                          SolutionWriter --- OpenFOAM-style instants
|   |-- curation/                    BESIDE the stack, not in it: a tool may read
|   |                                 the runtime, the runtime may read no tool
|   `-- applications/                choupoSolve / choupoBatch / choupoCtrl /
|                                     choupoSemiContinuous / choupoProps --
|                                     one main.cpp per binary
|-- bin/                            runCase, runTests, listCases, curate/check_*
|-- tutorials/                      end-to-end cases under
|                                     steady/ unsteady/ ctrl/ batch/ props/ plant/
|                                     (one folder per DISCIPLINE, not per binary)
|-- gui/                            web GUI (see CLAUDE.md Sec. 12)
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`-- README.md
```

Key idea. Data lives under `data/standards/`, write-guarded. The engine *refuses* to write under `data/standards/` — adding a number there is a deliberate curation act, not a side effect of a run. WHERE a number belongs (an intrinsic field in `components/<name>.dat`, a pair file in a catalogue, or an operation's `constant/`) is decided by the **data-governance doctrine** (`docs/ai/data-doctrine.md`): the canonical reference for every curation question, summarised in §24.

Key idea. There is *one binary per problem class*, five in all — they share the same library under `src/{core,thermo,solver,...}` and differ only in their `applications/<name>/main.cpp`:

- `choupoSolve` — steady-state simulation, $F(\mathbf{x}) = 0$ (root-finding);
- `choupoBatch` — batch / recipe-driven, per-vessel ODE integration + time-triggered events;
- `choupoSemiContinuous` — transient flowsheet, NO control loop, $d\mathbf{Y}/dt = f(\mathbf{Y}, t)$ with continuous inlets and outlets (start-up, disturbance, fed-batch, feed & bleed); the fifth class, ruled 2026-09-20, sharing ONE driver (`src/dynamicDriver/`) with `choupoCtrl`;
- `choupoCtrl` — dynamic continuous + control, $d\mathbf{Y}/dt = f(\mathbf{Y}, \mathbf{u}, t)$ with PID/schedule controllers;
- `choupoProps` — property evaluation + parameter fitting (a distinct problem class: neither root-finding nor integration).

We split by *problem class*, never by numerical *strategy* (no `*Foam`-style proliferation): within each binary the strategies (Newton, Wegstein, RK4, Nelder–Mead, ...) are run-time selections via dicts, not compile-time ones.

3 Build system

```
make all              # release, -O2, build/linux64Gcc/
make MODE=debug all   # -O0 -g3, build/linux64GccDebug/
make print            # show resolved variables
make clean            # rm objs + binary for current MODE
make distclean        # rm entire build/
```

Sources are discovered with `find src -name '*.cpp'` — drop a new `.cpp` anywhere under `src/` and it picks up automatically. Header dependencies are auto-tracked via `-MMD -MP` (`.d` files alongside each `.o`).

Platform string is derived from `uname -s -m`. Currently only `linux64Gcc` is exercised; `darwin64Clang` and `linux64Clang` should "just work" with the same Makefile (modulo `<filesystem>` linking on older toolchains — see §17).

The WASM build (`make wasm`) is a sibling target — see `make/wasm.mk` and `CLAUDE.md` §13 for the Emscripten quirks.

4 Build architecture: shared library, optional modules, case code

Since the build is *dynamic* and *modular*. Three forces drove it: incremental compilation (change one binary, don't rebuild the engine), a clean extension story (users add unit ops without touching `src/`), and — load-bearing for the modular architecture — making it hard to produce a standalone binary that carries the whole engine inside. The model: an engine built as a shared library, and small binaries that *reference* it.

> **Key idea.** A binary never embeds the engine; it links `libchoupo.so` at build time and finds it at run time via `rpath`. A binary copied to a machine without the library simply does not start. This is the same model — one engine, many small binaries that share it, so rebuilding a binary never rebuilds the engine.

4.1 The shared engine library (`libchoupo.so`)

`make lib` archives every engine object (everything under `src/` except the per-application entry points) into a single shared object:

```
make lib              # -> build/<plat>/libchoupo.so   ($(CXX) -shared)
```

The engine is compiled `-fPIC` (position-independent) on Linux. Each binary then links `-lchoupo` with `rpath` set to `$ORIGIN` — the binary's own directory — which is where the `.so` is built; the project-root symlinks resolve to the real binary location, so `$ORIGIN` still points at the `.so`. The result is tiny binaries:

Artefact	Size	Contains the engine?
`libchoupo.so`	1.7 MB	it *is* the engine
`choupoSolve` (binary)	86 KB	no — references it
`choupoCase` (a user's)	∼100 KB	no — references it

> **Common pitfall.** The `win64MinGW` cross-build keeps *static* linking on purpose: the generated `.exe` is self-contained. The `LIB_DEP` / `LIB_LINK` Makefile variables carry the platform branch (objects directly for `win64MinGW`, `-lchoupo` on Linux).

4.2 Optional modules — a clean modular split

The engine can carry OPTIONAL modules as separate `.so` files, so a slim core (the teaching engine) builds without them. The membrane module is Vítor's NF/RO research code — conceptually distinct from the core and kept in its own library so a core-only build is possible. Everything is GPL-3.0-or-later; the split is modular, not a licence boundary:

Library	Contents	Licence
`libchoupo.so`	core (thermo, solver, unit ops, reports)	GPL-3.0-or-later
`libchoupoMembrane.so`	NF/RO membrane (`spiralWoundModule`)	GPL-3.0-or-later

The mechanism is the same link-time stub-swap used for user code (§4.3), applied to a module:

1. The optional unit op lives under its own folder (`src/unitOperations/membrane/`); its objects are filtered *out* of `LIB_OBJS` and compiled into `libchoupoMembrane.so` instead.
2. It is **not** registered in `UnitOperation::registerBuiltins()` (the core must not depend on it). Instead a `registerMembrane.cpp` in the module defines a `registerModules()` hook that registers the type via the usual explicit factory.
3. `choupoSolve/main.cpp` calls `registerModules()` right after `registerBuiltins()`. The core binary links an empty `registerModulesStub.cpp` *only* when the module is absent; the Makefile (`MEMBRANE_PRESENT := $(wildcard src/unitOperations/membrane)`) links exactly the `.so` *or* the stub, never both.

```
# slim core-only build: remove the module source, rebuild
rm -rf src/unitOperations/membrane && make all
# -> only libchoupo.so is produced; the membrane type is unknown:
#    "unknown type 'spiralWoundModule'"
```

To add another optional module, do exactly what the membrane does: a folder under `src/`, kept out of `registerBuiltins()`, with its own `registerModules()` compiled into its own `.so`. (When a *second* module appears, generalise the single `registerModules()` hook into a small list — still explicit, still no auto-registration.)

4.3 Case-local code (`case/code/`)

A user — a student, a customer — may write their own C++ under a case's `code/` folder and have it compiled into a per-case binary. Build-time only: **no runtime `dlopen`, no macro self-registration**, faithful to the glass-box credo.

```
case/
  code/
    MyOp.H  MyOp.cpp        # the user's unit op (derives UnitOperation)
    registerUserTypes.cpp   # defines registerUserTypes() -> registerType("myOp",...)
    options                 # OPTIONAL: extra -I / -l / flags (see below)
  system/  constant/
```

What `code/` can register. The primary, tutorialised use is a new *unit operation*. But `registerUserTypes()` runs *after* every `registerBuiltins()`, so it can register **any type with an explicit factory** — the same hook, the same pattern:

What	via	Status
unit operation	`UnitOperation::registerType`	demonstrated (userOp01/02)
thermo model (γ, EoS, P^{sat}, c_p)	`ActivityModel::registerModel`, ...	possible, no tutorial yet
outer driver (sweep / optim / DesignSpec)	`OuterDriver::registerType`	possible
report / sizing / costing	`Report::registerType`, ...	possible

What the user writes. Three files in `code/`: the unit op (`MyOp.{H,cpp}`, a normal class deriving `UnitOperation` with a `solve()` override), and `registerUserTypes.cpp`, which puts the class in the factory:

```
// case/code/registerUserTypes.cpp
void registerUserTypes() {
    Choupo::UnitOperation::registerType(
        "myOp", [] { return std::make_unique<Choupo::MyOp>(); });
}
```

How it is compiled. `bin/buildCode` runs one ordinary `g++` (it prints the exact command — nothing hidden):

```
g++ -std=c++17 -O2 -I src \             # (1) the engine's headers
    build/<plat>/.../choupoSolve/main.o \  # (2) the engine main() (calls
                                           #     registerUserTypes)
    case/code/registerUserTypes.cpp \   # (3) the user's registration
    case/code/MyOp.cpp \                # (4) the user's unit op
    -Lbuild/<plat> -lchoupo [-lchoupoMembrane] \  # (5) engine.so (+ optional module)
    -Wl,-rpath,build/<plat> \           # (6) where to find the.so at run time
    -o case/code/.build/choupoCase     # (7) the per-case binary (~100 KB)
```

The user's `.cpp` are compiled fresh each time (their `.o` are temporaries in `/tmp`, dropped after linking — only the binary remains). The binary links the engine `.so` DYNAMICALLY (5): it references the engine, never embeds it.

The trick — who supplies `registerUserTypes()`. The engine's `main.o` *calls* `registerUserTypes()` (right after `registerBuiltins()`) but does *not* define it — an unresolved symbol. The linker resolves it from exactly one place, and that place is the build-time choice:

Binary	`registerUserTypes()` from	Result
stock `choupoSolve`	`registerUserTypesStub.cpp` (empty)	registers nothing
the case's `choupoCase`	`case/code/registerUserTypes.cpp`	registers `myOp`

In the `buildCode` command above the stub is *not* linked — only the user's file is — so the linker resolves the symbol with the user's definition. Plain symbol resolution, no weak symbols, no `dlopen`, no macros. (Optional modules use the same pattern with `registerModules()`, §4.2.)

How it is used — at run time.

```
choupoCase runs:
  main()
    -> UnitOperation::registerBuiltins()  // cstr, flash, mixer,...
    -> registerModules()                  // optional modules (membrane.so)
    -> registerUserTypes()   // THE USER'S: registry()["myOp"] = constructor
  Flowsheet reads  type myOp;  in flowsheetDict
    -> UnitOperation::New("myOp")         // look up in the std::map
    -> the user's constructor -> object -> the user's solve()
```

The factory is just a `std::map<string, function>`; `registerType` adds an entry, `New` looks one up. `myOp` sits beside the built-in types — no special case.

What `solve()` receives — streams and thermo. The op's whole world arrives through two arguments:

```
int solve(const DictPtr& dict, const ThermoPackage& thermo, int verbosity);
```

Streams (the `dict`). Choupo is sequential-modular: an op does NOT rummage a global stream registry. The `Flowsheet` *injects* the unit's inlet stream into the `dict` as sub-dicts, plus the unit's own `operation` block:

```
auto feed = dict->subDict("feed");        // F (kmol/s SI), T, P of the inlet
auto comp = dict->subDict("composition"); // mole fractions z_i
auto oper = dict->subDict("operation");   // YOUR parameters (from flowsheetDict)
```

You read your inputs, compute, and publish outputs into `produced_`; the Flowsheet binds them to the next units by the `outputs (... )` order. (For several inlets, the unit declares `inputs (a b )` and reads each.)

Thermodynamics (the `thermo`). The `ThermoPackage&` gives full access to whatever models the case configured — you do not need to know *which* (γ = NRTL? EoS = SRK?); you call the method and the selected model answers:

```
thermo.n();  thermo.indexOf("benzene");  thermo.comp(i);  // count, index, Component
thermo.comp(i).vp().Psat_Pa(T);                           // vapour pressure (Antoine/...)
thermo.Kvec(T, P, x, y);                                  // K-values (the case's gamma-phi)
thermo.activity().gamma(T, x);                            // activity coefficients gamma_i
thermo.Hliquid(T, x);  thermo.Hvapour(T, y);             // sensible-datum enthalpies
thermo.H_real(T, P, y);                                   // real gas (H_ig + EoS departure)
thermo.H_stream_formation(T, P, vf, z);                   // elements-datum (carries Hf)
```

So a user op can flash, balance energy, or solve an equilibrium with the case's thermo — exactly what the built-in CSTR / flash / column do; it is a first-class citizen, not an add-on.

> **Common pitfall. Transport properties (viscosity, conductivity, diffusivity) do not exist yet** — they are on the roadmap as a `TransportModel` with a factory. When they land they will be reachable through the same `thermo` handle; today a user op has thermodynamics only.

Command	What it does
`buildCode <case>`	*compiles only* (shows the exact g++ command, reports compiler errors); the development-loop command
`runCase <case>`	*runs* the case — and if it has a `code/` folder, compiles it first (visibly) via `buildCode`

Both step up automatically if invoked from inside the `code/` folder. External dependencies (a property library, GSL, Eigen, ...) go in an optional `code/options` file:

```
# case/code/options
INCLUDE = -I/opt/coolprop/include
LIBS    = -L/opt/coolprop/lib -lCoolProp
FLAGS   = -O3
```

Common pitfall. User code lives in `case/code/`, **never** in `src/`. Three reasons: isolation (the engine tree stays auditable), authorship / licence (`src/` is GPL-3.0-or-later; the user owns their `code/`), and pedagogy (a clear engine-vs-mine boundary). Those external libraries are the *user's* dependencies on the *user's* code — the engine itself stays dependency-free.

5 Core types

What you'll learn. Four types live in `src/core/` and `src/streams/` and propagate through every layer of the simulator: get familiar with them before touching anything else.

5.1 `Dictionary`

hierarchical plain text. Every case dict is parsed into a `Dictionary`. Key operations:

```
auto d = Dictionary::fromFile("system/flowsheetDict");

scalar  T = d->lookupScalar("T");                       // raw
scalar  P = d->lookupScalar("P", Dims::pressure);       // dim-checked
auto    sub = d->subDict("operation");
auto    units = d->lookupDictList("units");
auto    comps = d->lookupWordList("components");

// Path-based mutation --- used by outer drivers
d->setScalarAtPath("units[0].operation.refluxRatio", 3.2);
```

The tokenizer treats `[` and `]` as word chars (so paths work). Optional `FoamFile` headers are skipped silently for compatibility with files from other tools.

Three input forms for scalars. The parser accepts named-unit (`P 1 bar;`), the bracket (`A_w [-1 2 1 0 0] 1.5e-12;`), and raw SI (`R 0.5;`). Named and bracket forms tag the entry with its physical dimensions `[M L T Theta N]` so the dim-checked `lookupScalar(key, expectedDims)` overload can throw a clear error on mismatch. Raw entries pass unchecked — the caller is asserting the value is canonical SI.

Available named dimensions live under `Dims::` in `src/core/Dimensions.H` (`pressure`, `molarFlow`, `massFlow`, `temperature`, `length`, `area`, `volume`, `concentration`, `molarHeatCap`, `viscosity`, `diffusivity`, `permeabilityWater`, `massTransferCoeff`,...). Adopt them at the call sites where the unit op reads its operating parameters.

5.2 `ProcessStream`

What travels between unit operations (units: SI canonical):

```
struct ProcessStream {
    std::string         name;
    scalar              F;       // total FLUID molar flow [kmol/s]
    scalar              T;       // K
    scalar              P;       // Pa
    std::vector<scalar> z;       // FLUID mole fractions, size = NC
    scalar              vf;      // vapour fraction (VLE outputs)
    std::vector<scalar> s;       // solid molar flow per component [kmol/s]
    ParticleSizeDist    psd;     // size distribution of the solid phase
    std::string         category;// "" = process; non-empty = utility leg
};
```

Key idea. `F`, `z` and `vf` describe the *fluid* phases only; the **solid is carried alongside in `s[]`** (kmol/s per component, empty or all-zero = no solids), with its size distribution in `psd`. A crystalliser writes its crystals into `s[]`; the balance machinery (`src/reporting/BalanceMath.H`) reads `s[]` so mass *and* energy close over the solid too (see §25). The `category` tag (populated by `utility <name>;`) lets the energy ledger separate a utility leg — heat crossing the boundary — from a process stream.

Mass-basis flow is a derived view, not a stored field — if you need kg/s, call `F_mass(stream, thermo)` from `src/streams/StreamMass.H`, which computes $F \cdot \sum_i z_i \, \mathrm{MW}_i$ on demand. One source of truth for the flow (molar); mass cannot drift out of sync.

When you produce streams in your unit-op, fill *all* fields. At print sites you typically scale by 3600 for kmol/h and by 10^{-5} for bar, or hand the value plus its dimensions to `DisplayUnits::instance().format(v, dims)` which honours the case's `controlDict.units` preferences.

5.3 SimulationResult

What a single pass produces — input to post-processors and outer drivers:

```
struct SimulationResult {
    std::map<std::string, ProcessStream>     streams;
    std::map<std::string, std::map<std::string, scalar>> kpis;
    std::map<std::string, EquipmentSizing>   sizings;      // SizingPass
    std::map<std::string, CostBreakdown>     costs;        // CostingPass
    bool                                     converged;
};
```

The `kpis` map is keyed `unitName` → `{ kpiName` → `value }`. Outer drivers read it; post-processors append to `sizings` / `costs`. (The `sizings` field was named `dimensions` until; renamed because `Dimensions` is now the physical-dimensions tag.)

5.4 ThermoPackage

Holds:

- components (vector of `Component`),
- one or more `Phase` instances (vapor / liquid / solid),
- convenience getters: `nComponents()`, `component(i)`, `phase(name)`, `Psat(i, T)`, `gamma(phase, x, T)`, etc.

Every unit operation receives a `const ThermoPackage&` in `solve()`.

Two enthalpy datums — and why only one survives. The package exposes enthalpy on two references:

- `Hliquid(T,x)` / `Hvapour(T,y)` — the **sensible** datum, anchored at an arbitrary T_{ref}. Fine for a single-stream heat-up, useless across a reactor: it silently drops the heat of reaction.
- `H_stream_formation(T,P,vf,z)` — the **elements** datum (formation enthalpies at 298.15 K, the same reference as the ideal-gas H_{ig}). The reaction enthalpy then *emerges* as the difference of formation enthalpies, so a balance closes *through* a reactor, a crystalliser, or a heat of solution.

> **Key idea. One datum, no fallback (§25).** The energy-balance ledger runs on the elements datum *only*. The former silent "fall back to the sensible datum" arm was **deleted**: a component missing its `standardThermochemistry` now *throws*, naming the component, so a curation gap is loud, never a quietly wrong closure.

6 The factory pattern — explicit, no macros

> *What you'll learn.* The single registration mechanism used everywhere in the code, and why we refuse the macro-based alternative.

Every extensible base class follows this template:

```
class UnitOperation {
public:
    using Factory = std::function<std::unique_ptr<UnitOperation>()>;

    static void registerType(const std::string& name, Factory f);
    static std::unique_ptr<UnitOperation> New(const std::string& type);
    static std::vector<std::string> availableTypes();
    static void registerBuiltins();          // <-- called in main()

    // ---- virtual interface ----
    virtual const std::string& type() const = 0;
    virtual int  solve(const DictPtr&, const ThermoPackage&, int verbosity) = 0;
    virtual std::vector<ProcessStream> producedStreams() const { return {}; }
    virtual std::map<std::string, scalar> kpis() const { return {}; }

protected:
    // helpers...
};
```

`registerBuiltins()` (in the corresponding `.cpp`) is the single place that *lists every type*:

```
void UnitOperation::registerBuiltins()
{
    auto reg = [](const std::string& n, Factory f) { registerType(n, f); };

    reg("isothermalFlash", []{ return std::make_unique<IsothermalFlash>(); });
    reg("flash",           []{ return std::make_unique<IsothermalFlash>(); });
    reg("adiabaticFlash",  []{ return std::make_unique<AdiabaticFlash>();  });
    reg("bubbleT",         []{ return std::make_unique<BubblePoint>();     });
    reg("dewT",            []{ return std::make_unique<DewPoint>();        });
    reg("cstr",            []{ return std::make_unique<CSTR>();            });
    reg("pfr",             []{ return std::make_unique<PFR>();             });
    reg("heatExchanger",   []{ return std::make_unique<HeatExchanger>();   });
    reg("HX",              []{ return std::make_unique<HeatExchanger>();   });
    reg("mixer",           []{ return std::make_unique<Mixer>();           });
    reg("splitter",        []{ return std::make_unique<Splitter>();        });
    reg("distillationColumn", []{ return std::make_unique<DistillationColumn>(); });
    reg("column",             []{ return std::make_unique<DistillationColumn>(); });
}
```

`main.cpp` calls each `registerBuiltins()` once at startup, in order. That is the *entire* registration story. No magic, no surprises.

> **Common pitfall.** Do *not* introduce `REGISTER_TYPE(...)` macros, anonymous-namespace static instances, or any pattern that runs registration as a side effect of TU loading. We have rejected this approach explicitly because the linker can drop unused TUs and the failure mode is silent.

7 Adding a new unit operation

> *What you'll learn.* The five moving parts in outline — skeleton, implementation, registration, tutorial, regression. The full *tim-tim-por-tim-tim* walkthrough, with a real worked example and the mandatory WASM rebuild, is in §26.1; this is the at-a-glance version.

7.1 Step 1 — Skeleton

Create `src/unitOperations/<category>/MyOp.H`:

```
#pragma once
#include "core/Dictionary.H"
#include "thermo/ThermoPackage.H"
#include "unitOperations/UnitOperation.H"

namespace Choupo {

class MyOp : public UnitOperation {
public:
    MyOp() = default;
    const std::string& type() const override { return type_; }

    int solve(const DictPtr&         dict,
              const ThermoPackage&   thermo,
              int                    verbosity) override;

    std::vector<ProcessStream> producedStreams() const override { return produced_; }
    std::map<std::string, scalar> kpis() const override { return kpis_; }

private:
    inline static const std::string type_ = "myOp";
    std::vector<ProcessStream>      produced_;
    std::map<std::string, scalar>   kpis_;
};

} // namespace Choupo
```

7.2 Step 2 — Implementation

`src/unitOperations/<category>/MyOp.cpp` — implement `solve()`. Boilerplate to follow:

```
int MyOp::solve(const DictPtr& dict, const ThermoPackage& thermo, int verbosity)
{
    // 1. The Flowsheet INJECTS the inlet + your params as sub-dicts of 'dict'
    //    (sequential-modular: no global stream registry to rummage).
    auto feed = dict->subDict("feed");          // F (kmol/s SI), Tfeed, Pfeed
    auto comp = dict->subDict("composition");   // mole fractions z_i
    auto op   = dict->subDict("operation");     // YOUR parameters
    const scalar F     = feed->lookupScalar("F",     Dims::molarFlow);
    const scalar Tfeed = feed->lookupScalar("Tfeed", Dims::temperature);
    const scalar T     = op->lookupScalar("T",       Dims::temperature);

    // 2. Echo inputs (verbosity >= 3 --- the pedagogical default)
    if (verbosity >= 3)
        std::cout << "  [MyOp] T = " << T << " K, F_in = "
                  << (F * 3600.0) << " kmol/h\n";   // print in kmol/h, store SI

    // 3. Solve the model --- use solver/ classes for Newton/Wegstein.
    //    newton1D(f, dfdx, x0, NROptions) -> NRResult { root, iterations, converged }
    solver::NROptions nro;  nro.lower = 200.0;  nro.upper = 700.0;
    auto r = solver::newton1D([&](scalar x){ /* g(x) */ return 0.0; },
                              [&](scalar x){ /* g'(x) */ return 1.0; },
```

```
                            /*x0=*/Tfeed, nro);

    // 4. Build produced streams into produced_ (order = your contract;
    //    the Flowsheet binds them to names via 'outputs (... )')
    produced_.clear();
    ProcessStream out;
    out.name = "myOp_out";
    out.F = F;  out.T = T;  out.P = feed->lookupScalar("Pfeed", Dims::pressure);
    out.z.resize(thermo.n());                    // fill ALL fields
    produced_.push_back(out);

    // 5. Populate KPIs (they feed sizing / costing / sensitivity)
    kpis_.clear();
    kpis_["T_out"] = T;
    kpis_["F"]     = F;

    return 0;     // 0 == converged
}
```

7.3 Step 3 — Register

Edit `src/unitOperations/UnitOperation.cpp`:

```
#include "<category>/MyOp.H"
//...
void UnitOperation::registerBuiltins() {
    //...
    reg("myOp", []{ return std::make_unique<MyOp>(); });
}
```

7.4 Step 4 — Tutorial

Create `tutorials/myOp01_quick_smoke/` with `system/controlDict`, `system/flowsheetDict`, and `constant/thermoPhysPropDict`. Keep the case trivial so it doubles as a regression test.

7.5 Step 5 — Build, WASM, regression

```
make all                # recursive find picks the new .cpp up automatically
make wasm-gui           # MANDATORY: the GUI's WASM is a SEPARATE build (see below)
bin/runTests            # NaN/inf guard on every case + golden-master KPIs
```

Common pitfall. Rebuild the WASM or the browser will not see your type. The WASM solver is a separate compile from the native binary. Symptom of a stale `.wasm`: the native run works, but the GUI errors `UnitOperation::New: unknown type 'myOp'`. Fix with `make wasm-gui` — the incremental build tracks sources, flags and the standards tree by content, so a changed input always rebuilds; if it ever says "Nothing to be done" after a real change, that is a dependency-graph bug to report, and `make wasm-clean` is a deliberate diagnostic, not a ritual. Never validate with a bare `choupoSolve > /dev/null && PASS` loop — a solver can return NaN with exit code 0; `bin/runTests` is the only honest gate (§16).

All previously-passing tutorials *must* still pass.

8 Declaring what a unit does to material

Some engine machinery needs to know not what a unit *is* but what it *does to the material passing through it*. The carried species inventory is the current example: when a stream arrives with a `speciation {}` block and the unit resolves no chemistry of its own, the block has to cross somehow, and how depends entirely on the unit's material mapping.

A splitter is the easy case and the instructive one. It divides matter by *one* fraction common to every species, so a carried species inventory — which is extensive — divides by exactly that fraction. Algebra, not chemistry. A separator, membrane, filter or adsorber has no such fraction: its outlets are not proportional copies of its inlet, and the inventory cannot be divided without a model.

The distinction is **declared**, on the unit:

```
// UnitOperation.H -- the default is silence
virtual std::string materialMapping() const { return ""; }

// Splitter.H -- the capability, stated
std::string materialMapping() const override
{ return "proportionalExtensiveSplit"; }
```

> **Key idea. The name of the class is irrelevant; what is declared is what the contract reads.** It would have been shorter to test whether the unit is a `Splitter` — and it would have been wrong the day somebody wrote a selective splitter, or a proportional divider that is not called one. A capability that a caller must rely on is declared by the class that has it, never inferred by the caller from a name.

Two consequences worth stating, because the first instinct on both was the wrong one:

- **Refusing was considered and rejected.** The engine already knows the fraction; refusing would force an author to restructure a flowsheet around a limitation that does not exist.
- **So was falling back to re-derivation.** Letting the post-solve pass rebuild the two branch blocks from scratch would replace known arithmetic with a model-dependent answer — worse than either alternative.

What must close, and is checked: the two branch inventories sum back to the inlet's, species by species, and each branch stays electroneutral, because one common fraction cannot break a balance that held before it. `basis01_two_unit_chain` exercises exactly this on the far side of a model boundary.

When you add a unit whose outlets *are* proportional copies of its inlet, declare the mapping. When they are not, say nothing — the default silence is the honest answer, and the machinery that needs the capability will decline rather than guess.

9 Adding a new thermodynamic model

Same explicit-factory spirit as everywhere else, with `ActivityModel` / `EquationOfState` / `VaporPressureModel` / `HeatCapacityModel` / `Phase` as the base. The one difference from `UnitOperation` is **the factory takes construction arguments** (a thermo model needs the configuring sub-dict, and usually the component list), so its `Factory` signature is not the nullary unit-op one — match the base you are extending:

Base	`Factory` / register call
`VaporPressureModel`	`(const DictPtr&); registerModel(name, f)`
`HeatCapacityModel`	`(const DictPtr&); registerModel(name, f)`
`ActivityModel`	`(const DictPtr&, const std::vector<std::string>&); registerModel(name, f)`
`EquationOfState`	`(const DictPtr&, const std::vector<Component>&); registerModel(name, f)`

The detailed, copy-pasteable recipe — with the validation-first AAD step and the EoS data-doctrine path — is in §26.2. The short version, adding UNIQUAC as an activity model:

1. `src/thermo/activityCoefficient/UNIQUAC.{H,cpp}` inheriting `ActivityModel`. Override the pure virtuals `gamma(scalar T_K, const sVector& x)`, `n()`, `modelName()`; a constructor `UNIQUAC(const DictPtr&, const std::vector<std::string>&)` reads the pair parameters.
2. In `src/thermo/activityCoefficient/ActivityModel.cpp`, add the `#include` and *one* line in `registerBuiltins()`:

```
#include "UNIQUAC.H"
//...
void ActivityModel::registerBuiltins() {
    registerModel("ideal",  [](const DictPtr&, const std::vector<std::string>& n)
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<IdealSolution>(n.size()); });
    registerModel("NRTL",   [](const DictPtr& d, const std::vector<std::string>& n)
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<NRTL>(d, n); });
    registerModel("Wilson", [](const DictPtr& d, const std::vector<std::string>& n)
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<Wilson>(d, n); });
    registerModel("UNIQUAC",[](const DictPtr& d, const std::vector<std::string>& n)   // new
        -> std::unique_ptr<ActivityModel>
        { return std::make_unique<UNIQUAC>(d, n); });
}
```

3. Use it in a tutorial's `thermoPhysPropDict` (inside `equilibrium.liquid`):

```
activityModel { model UNIQUAC; binaryParameters {... } }
```

10 Property operations (choupoProps)

choupoProps runs *property operations* rather than a flowsheet. Each inherits PropertyOperation (same explicit factory as everywhere else; registerBuiltins() is called from src/propertyOps/PropertyOperation.cpp). An op reads a state (T, P, composition), a properties list and an output block, and writes a CSV the GUI auto-plots.

Which ops exist is not listed here. This paragraph used to carry seven names and the engine registers twenty-seven; a list of built-ins in a manual is a second home for a fact the code already states, and it drifts the first time somebody adds one. The authority is gui/schemas/operations/, and the props guide's operations chapter is *rendered* from it (bin/curate/props_ops_reference.py, with a runTests gate that refuses a stale render). The props–unit partition is derived too: an op is a unit op precisely when one of the three UnitOperation registries registers it.

Property keywords live in PropertyEvaluator.cpp: per-component Psat_<c>, gamma_<c>, y_eq_<c>, Cp_liquid_<c> (each guarded — e.g. Psat on a component with no vapour-pressure model throws a clear error instead of dereferencing null), and mixture scalars Z, v_molar, H_real, S_real, transport, etc. PropertyScan1D wraps each evaluation in try/catch, writing nan on failure so one undefined component never aborts the whole scan.

propertyScanTernary — ternary phase diagrams by reuse. Walks the composition simplex ($x_1 + x_2 + x_3 = 1$) at fixed (T, P) and, at every node, calls the SAME solver the isothermalFlash unit op calls — so the diagram cannot disagree with a single flash. Three modes:

- bubbleT — a boiling-temperature *surface*: per node BubblePoint::compute; needs only the three vapour pressures (ideal binaries are fine), no liquid-liquid data.
- lle (default) — a *solubility* map: the liquid–liquid flash (PhaseSet::LL) gives a clean binodal + tie-lines, no spurious subcooled vapour (the extraction/decanter view).
- vlle — the full vapour + two-liquid classification (PhaseSet::VLLE).

Each simplex node is independent, so the scan is embarrassingly parallel: the optional shard { k; n; } block computes only rows with $i \bmod n = k$ (round-robin, balanced). The GUI fans $N =$ cores-2 shards across N WASM workers and merges the partial CSVs (the union of shards reproduces the full grid exactly). No physics is reimplemented — the op is a triangular loop + a classifier + a CSV writer over solveCore/BubblePoint.

Adding a property op. Mirror the recipe of §6: implement src/propertyOps/MyOp.{H,cpp} inheriting PropertyOperation (override type(), run(dict,thermo,verbosity), optionally diagnostics()), then one reg("myOp", ...) line in PropertyOperation::registerBuiltins(). Rebuild the native binary AND the WASM (make wasm-gui) or the browser will not see the new op.

And write the schema — gui/schemas/operations/myOp.schema.json, with a title and a description for the op and for every key it reads. It is not optional decoration: the GUI builds its form from it, check_schema_coverage refuses a key the corpus uses and the schema does not declare, and the props guide's reference chapter is generated from it. An op without a schema is an op nobody can find.

11 The thermophysical property architecture

Property data in Choupo is *declarative systems with OpenFOAM-like explicit files*. A case does not hand the solver a bag of numbers; it *declares* its thermophysical system — `constant/thermoPhysPropDict` (`recordType thermophysicalPropertySystem; schemaVersion 2;`), one record naming the components, the equilibrium formulation, the model in each phase slot, and the parameter files the run needs. The one-sentence contract:

> *`thermoPhysPropDict` declares, `ThermoPackageBuilder` assembles, `ThermoPackage` computes; curation produces `.dat`, runtime does not estimate.*

11.1 The flow

```
thermoPhysPropDict        the case DECLARES this (declarative)
     |
ThermoPackageBuilder      runtime ASSEMBLY: loads + assembles, never estimates
     |                    (src/thermo/ThermoPackageBuilder.cpp)
ThermoPackage             the mounted runtime COMPUTE object (unchanged)
     |
unit operations           consume the ThermoPackage, as before
```

Terminology (load-bearing). The runtime assembly step is the `ThermoPackageBuilder` — it **loads**. It is *never* a "resolver": *resolver* is reserved for **curation-time** estimation (the Props Guide). Builder = runtime, loads; resolver = curation, may estimate; the two never mix.

11.2 The two axes

A substance never *is* molecular or electrolyte — the same NaCl plays different roles, and the role is chosen by the package a case selects, never stored on the substance. Two orthogonal choices define the role:

Representation (the *package* activates it): lumped → complete dissociation → partial / ion-pair → multi-ion. The complete, K-free split is the `dissociatesTo` ion stoichiometry on the component — a formula-like *identity*; the moment a split carries an equilibrium constant (an ion pair, a hydrate) it is *chemistry* and lives under `chemistry/`.

Reference (the *method* declares it): which datum the species anchor to — solid, pure-liquid Raoult, aqueous infinite-dilution, or fused-salt. `species/` records are medium-agnostic (one Na^+ file); the thermo data inside are medium-tagged blocks (`aqueousThermo`).

11.3 The data homes

`ls data/standards/` is the authority on this list, and that sentence is not a formality: this subsection was headed "the seven data homes" while the tree held eight directories, and the one it left out was `conventions/` — the versioned convention profiles the whole equilibrium-parameterisation identity references. A count written into prose is a derived number in a second home; it drifts, and it drifts silently. What follows is the *shape*, never the tally:

```
data/standards/
  components/<name>.dat        identity (name, MW, nonvolatile)
                               + dissociatesTo { Na 1; Cl 1; }  [salts]
                               + solidPhases{}: rho_p, k_v and the salt's
                                 OWN dissolution anchor (the retired
                                 phases/solid/<phase>.dat is a legacy read)
  species/<name>.dat           one modelSpecies file per species:
                               formula + charge + medium-tagged thermo
                               (charge 0 for O2/N2(aq); never fed to a flowsheet)
  chemistry/<name>.dat         FLAT: REAL equilibria (stoichiometry + K + dH);
                               each record's recordType names its family
                               (aqueousSpeciation / gasLiquidEquilibrium /
                               ionExchangeEquilibrium) -- never a subfolder
  parameters/
    Pitzer/{pairs,lambda,psi}/...          Pitzer binaries + ternaries
    eNRTL/<c>-<a>.dat                      eNRTL tau pairs
    NRTL/ UNIQUAC/ Wilson/ Henry/ SRK/ ...  molecular pair catalogues
  conventions/<profile>.dat    the VERSIONED, immutable convention profiles
                               a curated parameterisation cites by name
```

```
                              (Sander-Hxp-v1, PHREEQC-gasMolal-v1, ...):
                              what the constant MEANS, kept apart from
                              what it IS
  assets/<name>.dat           FLAT with a 'kind' field -- construction
                              materials, membranes, adsorbents, resins:
                              one home, filtered by kind, not by directory
  mixtures/                   curated mixture records
  utilities/<name>.dat        steam LP/MP/HP, cooling + chilled water,
                              dowthermA, hitecSalt, refrigerationPG,
                              electricity
  (the property SYSTEM is not a directory at all: a case DECLARES it INLINE
   in constant/thermoPhysPropDict -- THE CENTRE.  There is no shared package
   catalogue, and the old per-model methods/<name>.dat ceremony records were
   retired.)
```

`components/` stays *flat* (the loader resolves `<name>.dat` by exact name, O(1)). There is deliberately **no** second component record per salt (no `apparent/` twin — one component, one file), no user-facing "true species" vocabulary (an ion is not *more real* than the salt; it is the speciation a model postulates — so: *model species*), and no `propertySets/` kind (it had zero readers and was removed rather than kept as ceremony). A minimal electrolyte system (active chemistry beside it in `constant/chemistryDict`):

```
recordType    thermophysicalPropertySystem;
schemaVersion 2;
components ( water NaCl );
equilibrium
{
    formulation electrolyteGammaPhi;
    aqueous
    {
        solvent            water;
        activityModel { model Pitzer; }
        compositionBasis molality;
    }
    vapour { fugacityModel idealGas; }
}
```

The solid equilibrium activates through the chemistry selection (the K_{sp} anchor comes from the component's `solidPhases`); the Pitzer pair is resolved by cation–anion name from the parameter catalogue.

11.4 The builder contract

`ThermoPackageBuilder::build` dispatches on the declared `equilibrium.formulation`. For an electrolyte formulation it (1) reads the component list; (2) identifies the ACTIVE salt among the components whose record carries `dissociatesTo`: the only one if there is one, otherwise the one whose own `solidPhases{}` block owns the phase `constant/chemistryDict` declares — never the first in the list, and with neither it refuses naming every candidate — announces the choice (`[electrolyte] active salt: ...`), and takes its identity (MW, role) from `components/<salt>.dat`; every other salt rides as an ideal spectator; (3) classifies the ions by charge sign from each ion's `species/<name>.dat`; (4) loads the crystal (ρ_p, k_v) and the saturation anchor from the salt component's own `solidPhases{}` block, by that declared phase name (the retired `phases/solid/<phase>.dat` and `chemistry/salts/<phase>.dat` stay readable, announced as `[legacy]`, for old external cases only); (5) assembles the Pitzer/eNRTL kernel through the same `SaltFromCatalogue` helpers the legacy path used — byte-identical by construction; (6) mounts the `ThermoPackage`. **It loads and assembles; it does not estimate** — a missing record fails clear:

```
FATAL: electrolyteGammaPhi selected the Pitzer activity model,
       but required parameter pair Na-Cl was not found.
```

For a molecular formulation (`gammaPhi` with NRTL) the same entry point reads the binary pair from `parameters/NRTL/` and mounts the classical NRTL package — one stack, electrolyte and molecular alike. The reference conventions are the `standardState` keys on the phase/group blocks (`pureLiquid`, `infiniteDilution`); the old per-model `methods/<name>.dat` ceremony records were retired once the dispatch became fully native.

11.5 Status

The legacy salt-level electrolyte front-end is fully retired: the `electrolyte{}` component block is gone from every salt `.dat` (ion identity now derives from `dissociatesTo`; dissolution data live in the salt's own `solidPhases{}` block), `ElectrolyteActivity::configure()` and the `model pitzer;`/`model eNRTL;` factory registrations are deleted, and the electrolyte tutorials declare `electrolyteGammaPhi` systems. A molecular case is the same record degenerate — a `gammaPhi` block, no dissociation, no chemistry; lightweight is not legacy. Two electrolyte front-ends remain deliberately separate: *multi-ion speciation* (`pitzerHMW`/`davies`/`edwardsPitzer` via a `propsDict speciate` op — see §11.6) and the flat membrane/DSPM-DE data path. The mixed-solvent, segment-based eNRTL of Chen & Song (2004) ships as `ENRTLMixedSolvent` with a primary-data validation tutorial (`tutorials/props/electrolyte/enrtl_mixed_nacl_ethanol_esteso`).

11.6 Adding an AQUEOUS activity model — and the four things that are not the registration line

`AqueousActivity` (the `speciate` path) has a *nullary* factory, unlike the `ActivityModel` bases above, so a model whose parameters depend on the run must assemble them inside `evaluate()`. Adding `edwardsPitzer` in August 2026 exercised every way that can go wrong, and the list is worth having before you start, because *four of the five defects were invisible to a compiling, self-checking, unit-tested model.*

1. **Register it, or it is not a capability.** The model sat in the tree for a day: it compiled, its closed-form self-checks against the source paper passed, and *no case could reach it*, because it was missing from `AqueousActivity::registerBuiltins()`. No unit-level test can see this. Write the gate so it runs the model *through the engine* on a real case.
2. **Add it to the declared-name table too.** A case selects a model by a declared word (`model EdwardsPitzer;`), which `src/propertyOps/CasePackage.H` maps to the factory key. Keep the accepted list and the refusal message reading *one* table: the if-chain that preceded it held those two facts in two places, where a model added to one and not the other refuses by naming itself unimplemented.
3. **An `ActivityResult` outlives the frame that built it.** Its closures are called by the solver long after `evaluate()` returns, so they must be *self-contained*. The Edwards kernel captured its names and molalities by value and its parameter set by reference; the asymmetry was harmless while every caller outlived its result, and segfaulted the moment an adapter built the model as a temporary. Capture by value.
4. **Say what *your* model's scope is.** The speciation banner was a two-way switch on one model name with everything else falling through to Davies' sentence — so a new model inherited a description of a model that was not running, and both halves of it were false (the quoted trust band belonged to Davies; the "a_w from $\phi = 1$" claim was contradicted by the run's own a_w). A banner keyed on one branch and defaulting for the rest is a lie waiting for its second author.
5. **Teach `bin/choupo-import` about your parameter home, in the same commit.** This is the one that can corrupt data rather than merely mislead. A sealed case is forbidden the installation catalogue, so a record home the importer's dependency closure does not know is a record home the sealed case *loses*. The first sealed Edwards case kept 9 of its 28 pair parameters — the ones that need no record at all — and still ran, still converged, and answered with 19 terms missing. Nothing refused.

> **Key idea. Running is not agreeing.** The importer used to validate a seal by re-running the staged case with the catalogue hidden and checking the exit code, and that is exactly the check the defect above passes. `validate_staged_agrees` now compares the staged run against the case's own `expected` golden and refuses with the likely cause named, installing nothing. A case with no golden yet keeps the exit-code check and is *told* its answer is unverified — an unverifiable claim announced is not the same as one quietly not made.

Where the parameters themselves live is a separate decision, and the doctrine is unchanged: *the tree never stores a derivative.* Edwards measured one family of interaction parameters and estimates the rest from it by four rules; only the measured tables are on disk, and the estimated pairs are derived at assembly time in `EdwardsCatalogue.cpp` and printed with the rule that produced each one. Keeping the estimation out of the kernel is what lets the kernel stay a pure function of its parameters, testable against the paper's own closed forms with no filesystem in sight.

12 Adding components, materials, reactions

These are *data-driven* — no C++ changes needed. WHERE each number belongs is not a matter of taste: it is decided by the data-governance doctrine (`docs/ai/data-doctrine.md`, summarised in §24). Read that before adding or moving a value.

12.1 Components

Choupo's component data cascades through tiers, lowest precedence to highest: `local < standard < case < sector < unit` (`local` is the private, gitignored `data/local/` tier; every use is announced `[local] UNVERIFIED`).

- `data/standards/components/<name>.dat` — the **curated central catalogue** shipped with the simulator. Carries only **arity-1 intrinsic** data: identity, `critical`, `gasIdeal`, `liquidPure`, `solid`, `transport`, the `standardThermochemistry` datum, and any model-named parameter block (`pcsaft{}`, `uniquac{}`, `cosmo{}`). Treat it as read-only — the engine *refuses* to write here; a change is a reviewed curation act. A **pair** number (NRTL τ, Henry, k_{ij}, a heat of solution) is **never** copied into the `.dat`; it lives in a catalogue, referenced by name.
- `<case>/constant/components/<name>.dat` (and the same at `<sector>/` and `<unit>/`) — **per-scope overlays** for sample-specific refinements (a particular powder's sorption isotherm, a laboratory measurement).

> **Key idea. The overlay merges block-by-block, NOT field-by-field.** An overlay carrying `solid { rho_p 1610; }` replaces the *entire* `solid{}` block — the standard's `k_v` is *lost*, not deep-merged. A reference-state block is the atomic unit of physical meaning; mixing a sample's `rho_p` with the catalogue's `k_v` would be a silent hidden hybrid. **The curator must copy the whole block they refine.** (This corrects the pre-doctrine "field-by-field" behaviour.)

The file format is identical in both tiers:

```
name    benzene;          // the record's own key is 'name', not 'component'
formula C6H6;
MW      78.114;           // kg/kmol  (the key is upper case)
Tc      562.05;           // K
Pc      48.95;            // bar
omega   0.2103;
Tb      353.24;           // K
HvapTb  30720;            // J/mol at Tb   (one word, no underscore)
Vliq    8.943e-5;         // m^3/mol

vaporPressure
{
    model         Antoine;
    // log10(Psat [bar]) = A - B / (T[K] + C):  the three sit in ONE list
    coefficients  (4.01814   1203.835   -53.226);
    Trange        (287.7  354.07);
}

idealGasHeatCapacity
{
    model         polynomial;
    coefficients  ( -31.368  0.4746  -3.107e-4  8.524e-8 );
    Trange        (200.0  1500.0);
}

liquidHeatCapacity
{
    model         polynomial;
    coefficients  ( 162.94  -0.3493  9.51e-4  0.0 );
    Trange        (278.0  500.0);
}

standardThermochemistry
{
    dHf_298     82930.0;        // J/mol  (NIST ideal-gas value)
```

```
    s_298         269.20;       // J/(mol*K)  (third-law absolute)
    referenceState idealGas;    // the standard state dHf_298 / s_298 are on
}
```

Source the values from DIPPR / Reid-Prausnitz-Poling and *cite the source in a top-of-file comment* so they stay auditable.

The `referenceState` field of `standardThermochemistry`. The reference state for formation enthalpy is documented in the Theory Guide (chapter on the elements datum). The `referenceState` keyword names the thermodynamic *standard state* `dHf_298` and `s_298` are tabulated on, so `Component::h_formation(T, targetPhase)` can build the right integration path through Hvap / Hfus jumps and the matching c_p leg. Three values are allowed — and they are the words below, not the bare phase names: the engine reads `referenceState`, and a key spelled `phase` here is read by nothing at all (§12.1):

`referenceState`	required data	typical species
`idealGas`	`idealGasHeatCapacity`, plus `HvapTb`+Tc+Tb if the liquid leg of any stream is needed	*the convention — all 45 standard volatiles + radicals; follows NIST / JANAF*
`pureLiquid`	`liquidHeatCapacity`; `HvapTb`+Tc+Tb if any stream goes through the gas leg	reserved for compounds whose Hf is published in the condensed datum (rare)
`pureSolid`	`liquidHeatCapacity` (used as the dissolved-solute approximation $c_p^{\text{solid}} \approx c_p^{\text{liq}}$)	crystalline non-volatiles whose Hf cannot be referenced to gas because the compound never vaporises (sucrose)

The default is `idealGas` (the NIST / JANAF convention) for the overwhelming majority of volatile species, but **a condensed-phase datum must declare its `referenceState` explicitly** so the datum is unambiguous at the source of truth. A `pureSolid` or `pureLiquid` declaration without the matching c_p model throws a clear error from `Component::h_formation` the first time a stream containing the component is evaluated.

> **Common pitfall. This guide used to teach `phase gas;` here, and the parser has never read that key.** `Component::readFromDict` reads `referenceState` and defaults it to `idealGas` when it is absent, so a record written the old way is not refused — it is read on the ideal-gas rung whatever the author meant. Measured on a copy of `crystalliser01_sugar` with sucrose's `referenceState pureSolid;` replaced by `phase solid;`: exit 0, the cooling duty moves from 271.7 to 264.5 kW and the heat of crystallisation collapses from 4377.5 to 0 J/mol, and the only thing the run says is an advisory asking for sucrose's T_c, T_b and `HvapTb` — an ideal-gas leg the author never asked for. Whether an unread key inside a component record should REFUSE or announce is a contract decision and is reserved; until it is taken, the guide is the only thing standing between a reader and that answer.

> **Common pitfall. A missing `standardThermochemistry` is now fatal, not papered over.** The energy ledger runs on the elements datum *only* (§25); the old "fall back to the sensible datum with a note" path was deleted. A component that appears in a stream the balance touches but carries no formation datum throws, naming the component — a loud curation gap, never a quietly wrong closure. Curate the eleven historical gaps (or the new component's datum) before relying on the balance.

Add the new component name to the case's `thermoPhysPropDict components (...)` list (or to a per-case `constant/components/<name>.dat` overlay if the data is sample-specific) and the new entry is picked up automatically — no recompile needed.

12.2 Materials

`data/standards/assets/<name>.dat`, a record that declares `kind constructionMaterial;` (the assets home is flat and holds membranes and adsorbents too; the kind is what selects it):

```
name                <name>;
kind                constructionMaterial;
density             7850;   // kg/m^3
F_M                 1.00;   // Turton material factor (carbonSteel = 1)
sigma_y             140;    // MPa -- the ALLOWABLE stress, used as is
maxT                700;    // K   -- compared with the hottest port
maxP                100;    // bar -- compared with pressureDesign
thermalConductivity 50;     // W/(m K), the exchanger wall
```

`sigma_y` is read in MPa and used DIRECTLY as the allowable stress S of the ASME thin-wall formula (`src/postProcessing/sizing/VesselMechanics.cpp`); the engine applies no safety factor, so a record must store an allowable there and never a yield strength, whatever the key's name suggests. `maxT` and `maxP` are announced against, never enforced (`src/postProcessing/sizing/MaterialLimits.H`). `MaterialRegistry::loadFrom()` scans the directory (and a case's own `constant/assets/`) on simulator start. No registration step needed.

12.3 Utilities

`data/standards/utilities/<name>.dat` catalogues a plant utility (steam header, cooling-water loop, hot-oil loop, refrigeration brine). The same registry pattern as materials and membranes: the `UtilityCatalogue::loadFrom(dataRoot)` call in each main slurps the directory at startup; the stream parser then resolves `utility <name>;` keys against it.

```
utility     steamLP;
tier        heating;            // heating | cooling
mechanism   condensation;       // condensation | sensible | evaporation

components  ( water );
state       saturatedVapour;
P           2.5 bar;
T_in        400.3 K;
T_out       400.3 K;             // condensate same T as steam

dutyPerKg   2186000.0;           // J/kg delivered  (Hvap_water at T_sat)
cost        12.0;                // EUR/GJ delivered
costYear    2022;
```

```
description "Low-pressure saturated steam header";
```

To add a new utility, drop a new `.dat` into the directory — no C++ change. When the utility loop carries a non-water fluid (oils, salts, brines), the `components` list points at a corresponding entry under `data/standards/components/`; the catalogue ships `dowthermA`, `hitecSalt` and `propyleneGlycol30` pseudo-components for that purpose. See the user guide for the case-author syntax `utility <name>;` and the `utilities` report that aggregates kg/h, MW, and EUR/h per category.

12.4 Reactions

In a case's `constant/reactions`:

```
reactions
{
    ethanol_to_water_test
    {
        stoichiometry
        {
            ethanol      -1.0;
            water        +1.0;
        }
        kinetics
        {
            type        powerLaw;
            A           1.0e6;        // s^-1
            E           80000.0;      // J/mol
            orders      { ethanol 1.0; }
        }
    }
}
```

Referenced from a CSTR / PFR by name (`reaction ethanol_to_water_test;`).

> **Key idea. The reactions library cascades UP, like the thermophysical system.** When a branch of a sectored plant (§22) is run *standalone*, its named-reaction library resolves by walking up the parent-folder chain (capped at six levels) exactly as the thermo package does, and **announces its origin** — `reactions library: loaded -- LOCAL` vs `inherited from <parent>/...`. Before this symmetry was restored, a sub-sector run alone died with "reaction not in constant/reactions" because only the local folder was consulted: the two cascades were asymmetric. No-silent-crutch: the cascade is loud.

12.5 `bin/choupo-thermoml` — the local ThermoML archive

The one ThermoML tool. The engine carries no network and compiles to WASM, so downloading lives here and the engine only reads the cache this tool builds — `thirdParty/thermoml/` (gitignored; `$CHOUPO_THERMOML_HOME` overrides it), never `data/local/`, which the loader resolves by name. Nothing from the archive enters the repository.

```
bin/choupo-thermoml sync [--index-only]          # download the official NIST archive
bin/choupo-thermoml index                        # build the local index over the cache
bin/choupo-thermoml search <query> [--online] [--raw-query Q] [--max-n N]
bin/choupo-thermoml extract <query> [--into DIR]  # Choupo datasets for a compound
bin/choupo-thermoml extract-vle <file> --into DIR [--block N] [--provenance measured|synthetic]
```

Refusals, each by name: a root element of neither known dialect; a required element absent (matching is by local element name, never by path); `extract` with no index, or a query nothing in the index matches; `search -online` when the archive is unreachable; and an `extract-vle` file carrying several interpretable bubble-point blocks, listed by index and requiring `-block N` — choosing is the curator's act. `index` lists the files it could not parse rather than dropping them.

12.6 `bin/choupo-curate` — evidence to dossier, promoting nothing

```
bin/choupo-curate <compound> [--case DIR] [--rule byDoiOrder] [--nFit N] [--maxAAD pct]
```

One command in a curator's order: resolve the compound in the local ThermoML index, extract every dataset Choupo can read into the case, *declare* the partition (`byDoiOrder`: datasets sorted by DOI, the first `nFit` fitted, the rest held out — stated before any residual is seen, recorded in the dossier), run `choupoProps`, report the held-out residuals, write the evidence record. Only `vapourPressure` and `heatCapacity` have a fit operation; other properties are `SKIPPED`, and with none left it exits 1. A single dataset makes the validation `REFUSED`, not split smaller. Without `-maxAAD` no verdict is claimed. Nothing is promoted: the dossier sits in `<case>/curation/`; admission to `data/standards/` is the separate act `bin/curate/promote-from-dossier`.

13 Adding an outer driver / post-processor

13.1 Outer driver

`src/outerDriver/MyDriver.{H,cpp}` inherits `OuterDriver`. Override:

```
class MyDriver : public OuterDriver {
public:
    void readFromDict(const DictPtr&) override;
    int  run() override;       // calls simulator_(flowsheetDict_) N times
};
```

The simulator functor and dict templates are injected by `main.cpp`:

```
auto driver = OuterDriver::New(outerDict);
driver->setSimulator     (simulate);          // SimulationResult(DictPtr)
driver->setFlowsheetDict (flowsheetDict);     // mutable flowsheet template
driver->setThermoDict    (thermoDict);        // mutable thermoPhysPropDict
driver->run();
```

The simulator closure captures the thermo dict by `shared_ptr`, so any in-place `setScalarAtPath()` the driver makes on `thermoDict_` is picked up on the next `simulator_()` call.

Inside `run()`, you typically:

1. Mutate `flowsheetDict_->setScalarAtPath(...)` to change a sweep variable, OR `thermoDict_->setScalarAtPath(...)` to change a model parameter (e.g. NRTL `a_ij`).
2. Call `simulator_(flowsheetDict_)` → get a `SimulationResult`.
3. Extract KPIs / stream fields, accumulate residuals / responses.
4. Decide next step (LM / Nelder-Mead / GA / ...).
5. Eventually report and return 0 (converged) or 1 (failed).

Two concrete examples to read first:

- `src/outerDriver/SweepDriver.{H,cpp}` (∼ 150 LOC) — 1-D sensitivity sweep.
- `src/outerDriver/OptimizationDriver.{H,cpp}` — Nelder–Mead over the same functor, with `DesignSpec` constraints; the one to read when your driver has to *decide* rather than merely traverse.

> **Common pitfall.** This list used to send you to `src/outerDriver/FitBinaryPair.{H,cpp}`. That driver was **retired**: `OuterDriver::registerBuiltins` still registers the name `fitBinaryPair` purely so that an old case gets a refusal that *names its replacement* (`choupoProps`' `fitParameters` op) instead of an unhelpful "unknown type". The file is gone; the guide went on citing it as one of two worked examples to read first. A manual that sends a reader to a file that is not there costs more than a manual that says nothing.

Register your driver in `OuterDriver::registerBuiltins()`.

13.2 Post-processor

`src/postProcessing/MyPass.{H,cpp}` inherits `PostProcessor`. Single method:

```
int run(SimulationResult&) override;
```

Append to `result.sizings` / `result.costs`, write CSVs, print to stdout. Register in `PostProcessor::registerBuiltins()`.

`PostProcessor::buildChain(postDict)` walks `postDict` top-level keys and instantiates one pass per recognised key. Add yours to the chain-builder in `PostProcessor.cpp`.

14 Adding a sizing model

`EquipmentSize` is a sub-factory used by `SizingPass`. Add `src/postProcessing/sizing/MyEquip.{H,cpp}` inheriting `EquipmentSize`:

```
class MyEquip : public EquipmentSize {
public:
    std::vector<EquipmentSizing> size(const std::string& unitName,
                        const SimulationResult& result,
                        const Material&  material,
                        const DictPtr&   designRules) const override;
};
```

Use `result.streams.at(...)` and `result.kpis.at(unitName).at(...)` to get the process data, plus `designRules` for the design specs (pressure design, F_M, joint efficiency, ...). Fill `EquipmentSizing`:

```
EquipmentSizing d;
d.unitName       = unitName;
d.equipmentType  = "myEquip";
d.material       = material.name;
d.set("V_R",    ..., "m3");    // canonical size
d.set("t_wall", ..., "m");
d.set("weight", ..., "kg");
return { d };                  // ONE item: leave d.equipmentTag empty
```

A vector, because one unit may realise several physical items. A flowsheet unit is a mathematical operation; design equipment is a physical object, and the two are not always one to one. A `distillationColumn` is a shell, a tray stack, a condenser, a reboiler and a reflux drum, and `ColumnSize` returns all five — each with its own `equipmentTag` (`"shell"`, `"condenser"`, ...), its own `equipmentType` (which selects the cost correlation) and therefore its own specification sheet under `design/<SECTOR>/<unit>/<tag>`. Every other sizer returns one element and leaves the tag EMPTY, which keeps `EquipmentSizing::itemId()` equal to the unit name and every surface unchanged. Returning an empty vector is not a legal answer: a sizer that cannot size its unit `throw`s, naming the unit and the remedy, and `SizingPass` catches it by name.

> **Common pitfall. One missing item must not cost the others.** `SizingPass` catches per UNIT, so a `throw` from a five-item sizer loses all five. Throw only for what makes the unit unsizable at all; for an item whose design rules the case did not declare, ANNOUNCE it by name on `AdvisoryLog` with the remedy and return the others.

> **Common pitfall. Declare the unit; never write `d.values[...]` directly.** `set()` is the one door, and it writes both the value and its unit. A value written past it reaches the reader with no unit, and the design sheet writer refuses it by name.
> The unit is a *named* unit — `m3`, `kW`, `bar`, `W/m2/K`, or `-` for a dimensionless ratio — spelled exactly as the dict grammar spells it, because it is printed into a file a student reads. That spelling is not cosmetic: `(` and `)` are not word characters, so `W/(m2.K)` does not survive the tokenizer and makes the whole sheet a file Choupo's own `Dictionary` refuses. It named the same unit at the same factor and shipped in every exchanger sheet for three days before the first reader of one found it. It is not the `[M L T Theta N]` exponent vector: that form exists in the grammar too, in a different slot, and would read as noise on a specification sheet.
> **Declare the unit the value is actually in, and do not convert.** Choupo's sizing values are *not* uniformly canonical SI — there are values in `kW` and in `bar` in the shipped sizers — and rebasing one to SI moves numbers in every golden that pins it. The unit travels so the reader knows; converting is a separate, deliberate act.

Register in `EquipmentSize::registerBuiltins()`.

The `Turton` costing model reads `d.equipmentType` to dispatch to the right correlation — so if your new equipment needs new cost coefficients, edit `src/postProcessing/costing/Turton.cpp` accordingly.

15 Coding conventions

15.1 Style

- C++ 17. `std::variant`, `std::optional`, `std::unique_ptr`, `std::filesystem` — yes. No raw `new` / `delete`. No `goto`.
- **Brace style:** Allman / open-brace on a new line for function definitions; Egyptian (`if (x) {`) for control flow.
- **Indent:** 4 spaces. No tabs anywhere.
- **Naming:** `snake_case` for locals, `camelCase` for methods, `PascalCase` for types, `UPPER_SNAKE` only for macros (we use exactly one: include guards via `#pragma once`).
- Compile cleanly with `-Wall -Wextra -Wpedantic` and fix anything that fires.

15.2 Includes

- Headers compile in isolation — every `.H` includes what it needs.
- Forward-declare aggressively in headers; pull full definitions only in `.cpp`.
- The build does not use precompiled headers.

15.3 Errors

- User-facing: throw `std::runtime_error` with a message a student can understand. Don't `assert` on user-data invariants.
- Programmer-facing (impossible state): `assert(...)` is fine.
- Never silently swallow errors. If a sub-step fails, surface it.

15.4 Comments

- Default to writing *no* comments. Identifiers should be obvious.
- Equations and references *deserve* a comment. Cite the source (Reid-Prausnitz-Poling page, DIPPR ID, Henley-Seader chapter).
- Top-of-file banner is preserved on every source — do not delete.

15.5 Header banner

Every `.H` and `.cpp` starts with:

```
/*---------------------------------------------------------------------------*\
|  Choupo                                                                     |
|                                                                             |
|  Open-source chemical process simulator                                     |
|                                                                             |
|  Copyright (C) 2026 Vítor Geraldes                                          |
|  SPDX-License-Identifier: GPL-3.0-or-later |
-------------------------------------------------------------------------------
Class
    Choupo::<ClassName>

Description
    <one-paragraph description>
\*---------------------------------------------------------------------------*/
```

Don't omit this on new files — it doubles as licence notice.

16 Regression testing

Regression is run with `bin/runTests`.

```
bin/runTests                 # every tutorial (5 binaries + buildCode cases)
bin/runTests <case>          # one case
bin/runTests <directory>     # every case under it -- the chunk
bin/runTests --witnesses     # one witness per execution class
bin/runTests --record <case> # refresh that case's golden-master 'expected'
```

The testing ladder — run what the change can reach. The authority is `docs/architecture/verification-and-va` §3a; this is the working summary. *Do not* reach for the full sweep by default:

1. **While editing** — the touched case and the `check_*` gates that own the thing you touched.
2. **A coherent change closed** — the witness of the affected class (`tutorials/WITNESSES`).
3. **A bounded slice closed** — `bin/runTests -witnesses`, every execution class, minutes. **This is the rung `main` advances on.**
4. **The release TAG, or a genuinely cross-cutting change** — the full `bin/runTests`, once, and *in chunks* (`bin/runTests tutorials/steady/flash`).

Key idea. The full regression is a confirmation, not a discovery tool. It is also long enough that it is often started and rarely finished — and a run that never finishes verifies nothing while blocking the work. What the ladder gives up is stated rather than glossed: interactions *between* slices are caught at the tag instead of at each advance (§3a of the V&V document records the four real defects that argument has already cost).

`runTests` does two checks, both stronger than a plain exit-code loop:

1. **NaN / inf guard — every case.** The structured result block is scanned for NaN or inf values; a hit FAILS the case. This is the check a bare `binary > /dev/null && PASS` loop *cannot* do: a solver can return NaN with exit code 0, and the old loop reported it green. That exact trap once hid a broken `membrane01` (NaN KPIs, exit 0) after a pressure-unit refactor.
2. **Reference KPIs — cases with an `expected` file.** Each line pins a KPI or stream value to its golden-master value with a relative tolerance; any deviation FAILS, printing `got X expected Y`.

Key idea. The `expected` file lives in the case directory. Format (`#` comments allowed):

```
# kind   name      key         expected        reltol
kpi      flash01   V_over_F    0.30398357314   1e-6
stream   vapor     F           0.00844398814   1e-6
```

`kind` names WHERE the number is read from: `kpi` (inside the named unit's KPIs), `stream` (inside the named stream), `diag` (inside a props op's `operationResults[].diagnostics`, `name` = the op), `aad` (the top-level `validation` overlay table, `name` = the dataset, `key` = `<model>.<property>.<statistic>`), `closure` (the top-level `energyClosures` model-boundary ledger, `name` = the unit), `utility` (the top-level `utilityAllocation` array, `key` = `<tier>.<utility>.<field>`, or `<tier>.<port>.unserved` for a duty no catalogue utility can meet and `<tier>.<port>.carried` for one a physical stream carries to another unit — two different facts that once wrote the same row), and `csv` (a sweep's answer, which lives in a CSV and emits no result block), and `verdict` (a props op's `operationResults[].curation`, `name` = the op) — which is the one kind that compares a *word* rather than a number, so its tolerance column must read `exact`. A sixth column `anchor` is a CLAIM, not a kind: a self-recorded row pins that the answer has not *moved*; an anchor row carries a published value and pins that it *agrees*, and `-record` copies it through verbatim. Pin the numbers worth guarding; `-record` writes them from the current run once you trust it. A case with no `expected` still gets the NaN guard.

A golden is two claims, and its tolerance column has three forms (C384). A plain number (`1e-4`) is a *relative* band; `abs:<v>` is an *absolute* one, $|a - e| \leq v$ in the row's own unit; `exact` compares a word. On the platform that *recorded* a golden it claims "this has not moved": the band written in the file, tight.

On *another* platform it claims "this reproduces as well as this kind of number can": the file passed through one table, `bin/curate/golden_bands.tsv`, in a temporary stream that is never written to the tree. The table's platform classes are applied only there, and never tighter than the file's own band: a *count* (`iter`, `*iterations`, `*evaluations`) and a *condition number* (`cond_JtJ`) take a relative band of `2.0`, because they are decided by comparisons of 10^{-6} on a max-norm objective and are chaotic in the last places; an optimiser's *objective values* (`chi2`, `rms`, `bubble_*`, `relErr_*`, `lle_*`) take `1e-2` and its *fitted parameters* `5e-2` (a hundred times, and five times the square root of, the least-squares stopping tolerance). The one class that is on disk is the closure *residual*, because round-off of a balance is absolute on any platform: 10^{-5} of `residual_denom_kW` for a `boundary` row (or $\max(10^{-3}\,\mathrm{kW}, 10^{-4}|e|)$ when none is published), $10^{-8}|Q_{kW}|$ of the unit for `H_closure_gap_kW`, 10^{-9} of the largest pinned energy term of the ledger for `energy_residual_kJ`, never below 10^{-9}. The mode is decided *per golden file* by a platform *fingerprint* (CPU, architecture, C library, compiler and the SIMD features glibc dispatches on, read from the running system by one function in `golden_bands.py`): the file's own `# platform <fingerprint>` comment line, else the platform declared in `bin/curate/recording_platform.txt`; `-record` writes that line when it runs anywhere but the declared platform. Equal to the running platform: strict. Otherwise cross, and the run says so in one line at its top and one in its tally, with the number of files. `CHOUPO_GOLDEN_MODE=cross` (or `strict`) forces a mode, to see locally what the CI runner sees. A row that still disagrees across platforms after its class band is a *finding* and goes in `bin/curate/platform_divergence_debt.tsv`, which may only shrink.

Most of the corpus ships an `expected`; the rest run under the NaN guard alone. Add one whenever a case's numbers are load-bearing — and count them yourself rather than trusting a number written here:

```
find tutorials -name expected | wc -l          # cases with a golden
find tutorials -name '*.cho'  | wc -l          # cases
```

This paragraph used to name four cases as the ones carrying a golden. It was written when that was true and was never revisited, and by 2026-08 the real figure was two orders of magnitude larger — so a contributor reading it concluded their case's numbers were probably *not* pinned when in fact the odds were the other way. A count copied into prose is a derived value with a second home, which is the one thing this project's data doctrine forbids; the shape belongs here, the number belongs to whatever recomputes it.

The curation gates. `runTests` does not only run cases: it also runs the `check_*` scripts under `bin/curate/`, which is where the project's settled contracts are actually enforced (one home per value, no unread dict key, no commercial competitor named in the manuals, a sealed case that cannot change physics on sealing, and so on). Each is independently runnable, states in its own output what it does *not* cover, and has been sabotage-verified — a gate nobody has ever seen fail is not evidence.

16.1 `bin/runTests -record-append`

The append-only half of `-record`. It refreshes nothing: every existing row keeps its value, tolerance and claim (anchors included); only the `(kind, name, key)` triples the golden lacks are appended from a fresh run (kinds `kpi`, `stream`, `diag`, `aad`, `closure`, `utility`, `verdict`; generated tolerance `1e-4`, except `verdict` rows, which are generated `exact` because they compare a word). A kind missing from that list is dropped in silence — which is how the first `verdict` append reported "refreshed" over an unchanged file. The point is the diff: a change git, a gate and a reader can verify line by line. A case with no golden is skipped (creating one is a recording decision, not a sweep side-effect), and a case marked `.expect-nonconvergence` or `.known-broken` is never recorded. It reports `+ N new row(s) appended`.

Three instruments that are deliberately NOT gates. Each needs something `bin/runTests` cannot guarantee — a browser, a served copy, a site that publishes asynchronously — and a gate that skips when it cannot run is permanently green exactly where it matters. Each exits `2` when it could not honestly run, never a silent skip.

16.2 `bin/checkGui`

The GUI visual-defect harness. The node/jsdom tests cannot see layout (every box is 0×0); this opens the real app in a real Chromium at a real viewport and asks the one question they cannot: for every visible interactive control, would a click at its centre actually reach it?

```
bin/checkGui                # walk the EduTools workspace at 1400x900
bin/checkGui --screenshots  # also save a PNG per page (gitignored)
bin/checkGui --keep-server  # leave a dev server this run started up
```

Exit `0` no covered control; `1` at least one control is covered; `2` REFUSED — no Chromium, no `node_modules`, no dev server, empty registry, or `node` not on `PATH`. Implementation `gui/tools/checkGui/`; it drives Chromium over the DevTools protocol with no new dependency and never runs `playwright install`.

16.3 bin/drive-app

Open the *delivered* app in a real browser, run a case, read every view as a student reads it, and report what it actually fetched. Built 2026-09-02, when the frozen release app passed every at-rest check and, on running a case, fetched the solver from the site root — visible only by driving it. For each case it navigates to `<base>?case=<id>`, presses run, walks every view, records every request the page and its workers made, and reports requests leaving the app's prefix, views that throw or go empty, disabled controls with their hint, console errors.

```
bin/drive-app --base http://127.0.0.1:5173/ --case steady/flash/flash01_benzene_toluene
bin/drive-app --first-path                      # every ‘tier tutorial;‘ case, dev server
bin/drive-app --mirror https://www.choupo.org/v2608/app/ /tmp/m --first-path
bin/drive-app --serve /tmp/m --prefix /v2608/app/ --case ...
bin/drive-app ... --screenshots                 # PNG per view (gitignored)
```

Exit `0` no finding, `1` findings, `2` REFUSED (no `node`, an unknown argument). It runs from `RELEASING.md`'s freeze step, the one moment it must.

16.4 bin/checkSitePublished

Is `www.choupo.org` serving what `main` has? No arguments: it prints local `HEAD` and `origin/main`, fetches `/buildinfo.json` from the site and compares the served commit. Exit `0` `CURRENT` (the site serves `origin/main`); `1` when the stamp is absent or `UNCONFIRMED`; `2` outside a git repository or on an unreadable stamp. Not a gate: publishing happens minutes after the push, so at test time the site is legitimately behind. It can prove currency but not staleness: a commit that changes no built file leaves the site byte-identical and the stamp behind, so an older stamp is `UNCONFIRMED`, never stale — settle it from the publish run, where a cancelled run is the real failure.

17 Debugging

17.1 Debug build

```
make MODE=debug all
gdb build/linux64GccDebug/choupoSolve
(gdb) r tutorials/steady/flash/flash01_benzene_toluene
```

The debug build leaves a separate symlink `choupoSolve.debug` so the release binary stays usable in parallel.

17.2 Verbosity

`controlDict { verbosity 4; }` enables debug-level prints — every Newton iteration, every K-value evaluation, every Wegstein step. For LL flashes that misconverge, watch for stagnation: $g(x)$ should shrink monotonically near the root.

17.3 Sanitizers

Edit `make/compiler.mk` for the debug branch:

```
ifeq ($(MODE),debug)
    CXXFLAGS += -fsanitize=address,undefined -fno-omit-frame-pointer
    LDFLAGS  += -fsanitize=address,undefined
endif
```

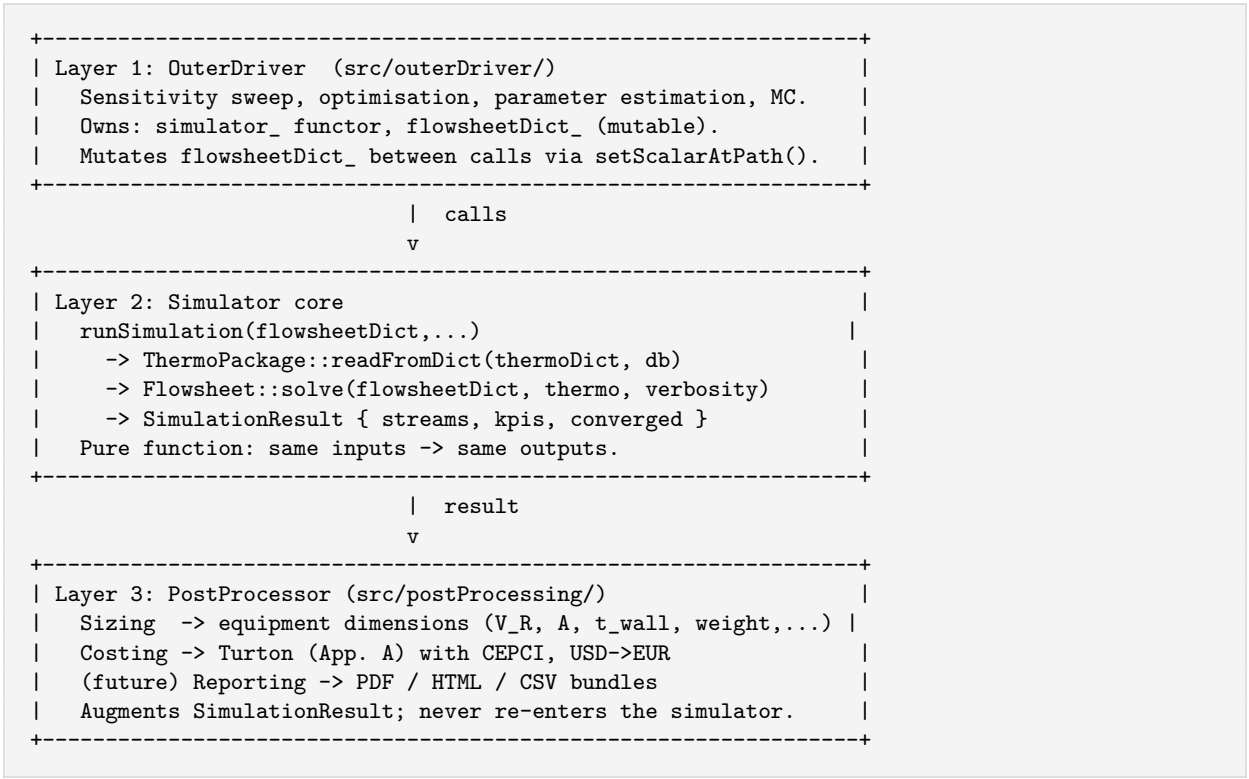
Then rebuild. Useful when a unit-op produces NaN out of nowhere or when adding a new factory and a dispatch crashes.

17.4 The most common failure: NaN propagation

Symptoms: `verbosity 3` shows clean inputs, then `V/F = nan` mid-flash. Almost always a $\log P_i^{\text{sat}}$ with $P_i \leq 0$ (Antoine outside its valid range) or an empty composition (`F = 0`). Add asserts where you suspect it.

18 Architecture deep-dive — the three layers

What you'll learn. The contract between the outer driver, the simulator core, and the post-processor. Read this twice before adding cross-layer logic.



The contract:

- Layer 1 *never touches the simulator's internal state* — it only mutates the dict it owns and calls the functor.
- Layer 2 *is pure* — `runSimulation(d) == runSimulation(d)` for the same dict. This is what lets Layer 1 do finite-difference Jacobians and Levenberg-Marquardt cleanly.
- Layer 3 *never re-runs the simulator* — it only reads the result. If you need to re-run (e.g. for utility costing), do it via an outer driver.

Key idea. Rule of thumb: if a quantity can be computed from a converged stream table plus KPIs, it belongs in a post-processor. If it requires re-solving the flowsheet, it belongs in an outer driver.

19 Module boundaries — the layering, and the one rule that keeps it

> *What you'll learn.* Which subsystem may include which. The three layers of §18 are about *control flow*; this section is about the *include graph*, and the two are not the same thing.

The three-layer picture above tells you who calls whom at run time. It does not tell you who may `#include` whom, and a project can obey the first while quietly violating the second for years. Choupo did. The layering below was declared in 2026-08-04 and is checked on every regression run.

19.1 The bands

`src/` is divided into subsystems, and the subsystems into five bands. A subsystem may depend **downward** (to a larger index) and **sideways** (within its own band). Upward is a violation.

Band	Subsystems	What lives there
0	`applications`	the four `main.cpp` files; nothing includes them
1	`outerDriver`, `postProcessing`, `reporting`	things that consume a finished run
2	`result`, `io`, `unitOperations`, `propertyOps`, `control`	the models and the records a run produces
3	`thermo`, `streams`, `materials`, `solver`	the physics and the numerics
4	`core`	`Dictionary`, `Types`, `Constants`, the plain records

19.2 The tooling plane

`curation` is *not* a band. It sits beside the stack under a rule that is stronger than any band:

> **Key idea.** A tooling subsystem may read the runtime. **Nothing in the runtime may read a tooling subsystem** — at any band, sideways included. Only `applications/` may join the two planes, because a binary's `main` is where a tool is assembled.

It was first placed in band 1, beside `reporting`, on the reasoning that it has reporting's shape: a consumer of the engine producing an artefact for a human. That is true, and it is the wrong conclusion. A band says "things at this level may reach sideways to you", which is exactly the permission a curation tool must never have.

19.3 Why it is worth a gate

A large open-source process simulator is the cautionary example, and it is instructive precisely because that project is otherwise competently built. (This manual names no other simulator; the architecture fact is what is worth carrying, and the product name is not.) Its contracts assembly has *zero* project references — a pure contracts layer at the bottom, which is the pattern `core` follows here. But its flowsheet solver references the diagnostics assembly in order to record anything, and that assembly holds its collector beside a window class that imports a desktop GUI framework. So that flowsheet solver has a compile-time path to a docking-panel GUI toolkit. Nobody chose that; it arrived by transitivity, one reasonable-looking include at a time.

Choupo had the same edge in miniature: `Flowsheet.cpp` included `reporting/ModelBoundaryAudit.H`, because the audit returned a record that lived above it.

19.4 How to place a new subsystem

`bin/curate/check_layering.py` refuses a subsystem it has no band for. It does **not** default one into the bottom band — that would put it where nothing can violate, which is a green light disguised as a check. So adding a directory under `src/` means declaring where it sits, in the gate and in `docs/architecture/module-boundaries.md`, in the same commit.

> **Common pitfall.** Ask what the thing *depends on*, not what it is *for*. A draft of the boundary document placed `io` beside `streams` because `SolutionWriter` "is the persistence face of the stream-state contract". True, and wrong: it also reads the whole `SimulationResult`, so that band makes `io` → `result` an upward edge. The narrative was about persistence; the include graph was about the result record, and **the include graph is what the compiler obeys.**

20 How a class is shaped — five rules, and what each one cost

> *What you'll learn.* The rules every model class in this tree obeys, each with the defect that paid for it. If you remember one sentence: *a fact has one home, that home is neutral, the engine can read it, derived things are derived, and a quantity carries its basis.*

These are not five rules. They are one rule seen from five sides, and they are stated here with their violations because a rule with a scar is remembered and a rule without one is negotiated.

20.1 Rule 1 — a fact has ONE home

A derived quantity is computed, never stored beside its inputs. *Trees never store derivatives.*

What it cost. `K_b`, the boiling-point-elevation constant, was stored in `water.dat` beside the `Tb`, `MW` and `HvapTb` that determine it by $K_b = RT_b^2 M/\Delta H_{\text{vap}}$. The copies had drifted: 0.512 stored against 0.512942 implied, 0.18 % apart, feeding straight into every evaporator's boiling-point elevation. A comment in the record said so, and nothing acted on it.

The same shape, four more times in one day: five hand-compiled counts wrong in both directions; waivers spread across eight of the gate scripts; the forward reaction `order` with five readers and five defaults, so a reactant with no declared order silently left its own rate law.

How to obey it. Derive, and keep any declared value as a *validating anchor* whose comparison the run announces. `Component::K_b()` is the worked example. If two places must know something, one of them computes it and the other asks.

20.2 Rule 2 — that home is NEUTRAL

Shared logic goes to the lowest band that can *own* the concept without acquiring upward dependencies — not merely the lowest band that compiles.

What it cost. Every upward edge in the subsystem graph was the same defect: one shared concept filed inside one of its two consumers.

The edge	The concept, and where it was hiding
`core` → `streams/thermo/unitOperations`	`FlatUnit` — four strings of topology — declared inside `SimulationResult`, so `core` inherited the result record's whole dependency set
`unitOperations` ↔ `propertyOps`	`readExchange`, a record reader that both a props bench and a process unit need, living in `namespace propertyOps`
`reporting` ↔ `postProcessing`	`OdsWriter`, a zip-based spreadsheet writer, filed under `reporting`
`unitOperations` → `reporting`	the finding *records* declared inside the result that carries them

20.3 Rule 3 — the ENGINE can read it

A fact that changes behaviour must live in a **parsed field**, never in a comment, a banner or a header.

What it cost. Three times in three days:

- 67 component records carried `PROPOSAL TIER -- UNVERIFIED` in a banner the parser discards — including 7 of the 16 in `cavett01`, the flagship external-reference case. Nothing announced them. `reviewStatus` is a parsed field now.
- `PolynomialCp` assigned `Tmin_`/`Tmax_` and never read them, so the commonest C_p path made no validity claim at all — which is why six inverted windows did no visible harm. *Harmless-because-unchecked is not safety.*
- `water.dat`'s formation datum is the ideal-gas one, and the fact was stated in a comment. Deriving the water-dissociation equilibrium constant from the records therefore gives $\log K_{25} = -12.4986$ against a stored -14 — a factor of 32 in K. With the liquid values the same arithmetic gives -13.998.

> **Key idea.** If you would write it in a comment to warn a reader, the engine needs it too.

20.4 Rule 4 — DERIVE, don't re-implement

One place computes each derived quantity, from the fundamental relation, for every combination of models — rather than each model carrying its own copy.

OpenFOAM is the reference implementation of this idea. It composes a thermo type from interchangeable layers and then derives everything else once:

```
g(p, T) { return this->ha(p, T) - T*this->s(p, T); }      // G = H - TS
K(p, T) { return exp(-this->Y()*this->gStd(T)/(RR*T)); }  // K = exp(-dGO/RT)
```

The equilibrium constant is not a model and not a correlation — it is an identity, computed once, valid for every equation of state crossed with every caloric model. That is where OpenFOAM's extensibility actually comes from: a new equation of state is a new class, and every derived quantity arrives free.

Common pitfall. Choupo is not there yet, and the guide says so rather than implying otherwise. K-values are the *primitive* here (`Kvec`, `stageK`), computed per model, and Gibbs minimisation lives in `gibbsMethod/{ElementPotential,DirectMin,ReactiveFlash}` plus the liquid–liquid path in `IsothermalFlash`. Phase equilibrium and chemical equilibrium are classically *one* condition applied to different degrees of freedom; here they are separate solvers. This is a known gap, not a settled contract, and its prerequisite is rule 5.

20.5 Rule 5 — a quantity carries its BASIS

When a value is expressed in one basis and a consumer needs another, the **conversion is an object**, not an assumption.

OpenFOAM solves this twice, identically, in unrelated parts of the code — which is the strongest evidence that it is structural rather than idiomatic. In the thermo layer, `gStd(T)` is an interface function every caloric model must supply, so a potential's reference state is part of the contract. In the particle layer, a distribution carries `Q_` (the moment order it is *declared* in), `sampleQ_` (what the consumer wants) and `q() = sampleQ_ - Q_`.

Choupo obeys it on the particle side: `src/core/distribution/` has Rosin–Rammler declaring the **mass** basis (a sieve), log-normal declaring **number** (a count), and `moment(k, basis)` doing the conversion in one place. Ask either for `d32()` and the answer is right without knowing which convention its author used.

On the thermo side it obeys it *at the record*: `standardThermochemistry { dHf_298; s_298; referenceState; }` names the standard state the numbers are on, and `h_formation(T, phase)` crosses rungs correctly, taking the transition at 298 K.

Common pitfall. It did not obey it *at the consumer* until 2026-08-06. `h_pure_ig`, `s_pure_ig` and therefore `g_pure_ig` read the two numbers straight off the record and integrated the ideal-gas C_p whatever was declared — and their callers are both Gibbs reactors, `Reaction::Kp`, `ReactiveFlash` and `H_ig`/`S_ig`. A `pureSolid` datum read there is wrong by a heat of sublimation: right sign, plausible magnitude, no diagnostic.
The catalogue could not reach it — every non-gas-rung record happens to lack a gas C_p, so the call threw. But it threw with *"needs idealGasHeatCapacity block"*, and a curator following that advice — which is reasonable for H_3PO_4 or HNO_3, both of which have good gas-phase data — would have converted an honest refusal into a silent wrong answer. **The error message was advice that creates the bug.**

20.6 The shape of a new class

Every physical model in this tree is the same shape:

```
class Thing
{
public:
    using Factory = std::function<std::unique_ptr<Thing>(...)>;
    static void registerType(const std::string& name, Factory f);
    static std::unique_ptr<Thing> New(...);
    static void registerBuiltins();          // called EXPLICITLY from main
    virtual ~Thing() = default;
    virtual std::string type() const = 0;
};
```

`ActivityModel`, `EquationOfState`, `HeatCapacityModel`, `VaporPressureModel`, `UnitOperation`, `OuterDriver`, `PostProcessor`, `CostingModel` — and `SizeDistribution`, which was a bare `struct` of two vectors until it was made to conform. That exception is worth noting because of what it cost: while the particle size distribution was a struct, no moment was computed anywhere in the tree and a continuous law had nowhere to live.

21 Gates — how a contract becomes enforceable

> *What you'll learn.* What a `check_*` script is, when your change needs one, and the two mistakes that make a gate worse than none.

A gate is a script under `bin/curate/` that `bin/runTests` runs and that exits non-zero when a structural contract is broken. There are more than ninety. They are not unit tests: a unit test checks a computed answer, a gate checks that a *rule* still holds across the whole tree.

21.1 The three-part test for a consolidated contract

1. the contract is **written**;
2. the engine **refuses** violators, by name, with a remedy;
3. a **case fires that refusal** — not a case that describes the structure, a case that exercises it.

The third is the one that usually goes missing, and a contract with only the first two is a preference.

21.2 Then sabotage it

Write the gate, then break the thing it guards and confirm it fails. This is not optional and it is not ceremony. Four times in one week a gate survived a sabotage it should have caught, and each time the *claim* was wrong rather than the code: an arm that tested a tabular sum while calling it a quadrature; a coverage check matching a substring of the very call it was meant to detect; an arm that measured truncation and called it discretisation.

> **Key idea.** A sabotage a gate survives is a claim the gate should stop making. Narrow the claim to what the sabotage showed it can see — a gate that reports less than it does is a nuisance; a gate that implies more than it checks is a hazard.

21.3 A check that cannot run must not pass

Found twice in one day. `bin/buildSite` warned and published when `pdftotext` was absent, so the guide-version check never ran in CI — the only place that publishes. And a gate reported `PASS` on every run while both of its inputs had been deleted. A permanently-green gate is worse than no gate, because it is counted.

So: if a gate's probe will not compile, or its input is missing, it must **fail**, not skip.

21.4 Where waivers live

Nowhere except `bin/curate/debt_registry.py`. Accepted violations were once scattered across eight gate scripts — the arity doctrine broken by the machinery built to enforce it. Each entry carries why, the remedy, and the blocker. `check_debt_registry` keeps it the only home and refuses an entry no gate reads.

21.5 Two chores that ride with a gate

- Adding a gate means regenerating `generated/gateManifest.json` (`bin/curate/gate_manifest.py`) in the same commit. The manifest records what each gate *claims*, derived by running it and capturing its own OK line — never transcribed from a docstring.
- Adding a tutorial case means regenerating `generated/releaseInventory.json` (`bin/curate/release_inventory.py`). Corpus counts are derived facts and have exactly one home — rule 1, applied to the project's own paperwork.

22 A plant and its sectors — nesting and `flattenNode`

> *What you'll learn.* How a plant of sectors of units collapses to ONE flat solver problem, why the hierarchy is namespace-only, and how a branch runs standalone.

A sectored Choupo case is **a plant and its sectors**: the plant nests composite *sectors*, and a sector nests leaf units. (The layout was formerly called "fractal"; the name was retired on 2026-10-09 because every case has those two levels and the word promised a recursion no case uses. `flattenNode` itself recurses and sets no depth limit, so the two levels are a convention of the corpus, not a refusal of the engine.) The solver never sees the tree. `flattenNode` walks the declared children and collapses the hierarchy into **one flat solver problem** with dotted `plant.sector.unit` names — the hierarchy is *authoring and namespace* only, never a recursive solver-within-a-solver. One flat Newton-on-tears recycle solves the whole plant.

Key idea. The hierarchy is discovered from the *dicts*, not the filesystem. `flattenNode` iterates the `children(... )` declared in `flowsheetDict`; a child loads as a sector only when its own `flowsheetDict` exists. This is why the numeric instant directories (§23) sitting beside the sector folders are never mistaken for sectors — they are not declared children and have no `flowsheetDict`.

Lean sector folders + the dual-reader. A node's topology may live either way, both first-class:

- the classic `<node>/system/flowsheetDict`, or
- the lean `<node>/flowsheetDict` directly at the node root.

A **dual-reader** keeps both loading: the child loader reads `<child>/flowsheetDict` first and *falls back* to `<child>/system/flowsheetDict`; the top-level `choupoSolve` does the same at the case root. Existing sectored cases (esterification, twoSectorDemo) load byte-identically through the fallback — **no forced mass-migration**.

A branch runs alone. Because both `thermoPhysPropDict` and the reactions library cascade UP the parent chain (§12), pointing `choupoSolve` at a sub-sector folder runs that branch as a complete case, inheriting the parent's thermo and reactions and announcing what it inherited.

The model-boundary rule. A per-unit `thermo {}` override REPLACES the global models for that unit (components stay global, via `thermoFor`). A stream crossing such a boundary is re-interpreted there. The governing law: H **is the conserved truth,** T **is the model-dependent readout**. A model boundary is not a physical device, so there is no real ΔT to absorb; H-continuity is a convention, not a law. The default is **hold-T, let-H-jump** (the discontinuity is visible in the printed enthalpy); a silent "hold H, flex T" is rejected. The honest opt-in is a **model-boundary audit** — print $\Delta H = H_{\text{down}} - H_{\text{up}}$ at fixed (T, P, z), fold it into the first-law ledger, and hard-refuse across any phase / `vf` / speciation flip. Full rationale: `docs/ai/energy.md`.

23 Solution instants — OpenFOAM-style per-iteration snapshots

> *What you'll learn.* Where the converged (and intermediate) stream tables are written, and why the layout is faithful to OpenFOAM's time directories.

A run writes **durable per-iteration snapshots** of every stream, modelled on OpenFOAM's time directories. `src/io/SolutionWriter` owns the writes; the durability core is atomic (`.tmp` → `rename` + `fsync`, gated on `flushEach`), a restart re-seeds the recycle tear from the latest instant, and a litter sweep purges stale numeric dirs.

The layout — instants at the CASE ROOT. An instant is a bare iteration-*number* directory at the case root, beside `constant/`, `system/` and the sector folders. "The time speaks for itself" — there is no `solution/` wrapper.

```
case/
  constant/  system/  reports/        # the usual case payload
  FERMENTATION/  CONCENTRATION/  ...   # sector folders (alphabetic)
  0/                                   # instant 0 (the initial seed)
    streams                              # ROOT flat stream table (the spine)
    FERMENTATION/ streams                # per-sector view
    CONCENTRATION/ streams
    byUnit/                              # per-unit breakout
  1/  2/  3/                           # later recycle iterations
  latest -> 3                          # convenience symlink to the highest
  solution.log                         # the instant index
```

Key idea. A stream IS a surface field (ϕ); the OpenFOAM analogy is exact. The payload file is named `streams`; `latest` is the highest-numbered instant (latestTime style — restart discovers it by the max integer dir name). A load-bearing *all-digit filter* on the numeric-dir scan guarantees a sector folder (alphabetic) is never swept or mistaken for an instant, and the sector loader iterating declared children (not a filesystem walk) guarantees the reverse.

The pilot is `tutorials/plant/ChemicalPlantTutorial`; `writeInterval` controls how many instants land (set it to `1` on a 3-iteration case or you only see `0` and the final — and miss the whole tear-residual march).

24 The data cascade — arity, scope, kind

What you'll learn. The mechanical decision procedure for WHERE every thermodynamic number lives. This is the engine-side summary; the canonical, ratified reference is `docs/ai/data-doctrine.md` — read it before any curation act.

Key idea. A datum's home is decided by ARITY, then SCOPE, then KIND — in that order. Read the quantity's *definition*, not its frequency of use.

(1) Arity — how many species does the definition name?

- **Arity-1** (the molecule alone on the table, no partner named — *not even implicitly*) → INTRINSIC, `data/standards/components/<n` T_c, P_c, ω, MW, `standardThermochemistry`, `solid`, `liquidPure`, `transport`.
- **Arity-2** (the definition must name a partner — a solvent, a counter-ion, an interaction species, *even water*) → PAIR/SET data in a **catalogue**, keyed by the pair, cited per value, referenced by name. NRTL τ, Henry H, Pitzer β, EoS k_{ij}, a heat of solution. **Never copied into the `.dat`.**

(2) Scope — at what level is the number TRUE? A datum lives at the *highest level where it is true* (the canonical value is a curation act in `data/standards/`). A lower node overlays only when it makes the value *more* true — a sample-measured refinement. Precedence, low to high: `local < standard < case < sector < unit` (`local` = the private, announced `data/local/` tier).

(3) Kind — the molecule, or the molecule-in-a-machine? A *rate* (kinetics) or a *geometry / distribution* (PSD) is NOT component data at all — it lives in the operation's `constant/` (`constant/crystallisation`, `constant/dryingKinetics`, `constant/reactions`). PSD is a *stream* attribute.

The aqueous TIER and the default-solvent resolver. Water gets *no* exemption from the arity test: an "in-water" property is pair data. Its one earned privilege is a single canonical, by-name aqueous reference tier — `data/standards/species/<ion>.dat` for ions, and `data/standards/solution/<solute>-<solvent>.dat` for molecular solutes (the heat of solution). The package declares a default solvent once (`solvent water;`); when an in-solvent property is needed and no solvent is named at the call site, the resolver uses the default **and announces it every run**:

```
[thermo] dHsoln(sucrose): solvent not named -> DEFAULT = water;
         solution/sucrose-water.dat  [Putnam & Kilday 1986]
```

Off-default it *fails with a remedy* ("provide solution/sucrose-ethanol.dat or set `solvent water;` explicitly") — it never silently substitutes the water number. This is implemented in `ThermoPackage` (`solventName_`, the `SolutionRegistry` walk).

The overlay merges block-by-block. As in §12: a reference-state block (`solid{}`, `gasIdeal{}`, `liquidPure{}`, each `eosParameters.<model>{}`) is the atomic unit of physical meaning. An overlay replaces the whole block — the curator copies the whole block they refine; a lone-scalar overlay is the forbidden hidden hybrid.

Model extensibility — `eosParameters{ <model>{...} }`. The same definition test, asked of the *model*:

1. A model needing only the corresponding-states triad $\{T_c, P_c, \omega\}$ (every cubic: SRK, PR, ...) **reads the triad off the Component and derives its own** $a_c, b, m, \alpha(T)$ **in source** — adding it touches *zero* data files (this is why adding PR after SRK touched no `.dat`). Do not pre-bake a_c into the `.dat`.
2. A model-specific PURE parameter that cannot be generated (PC-SAFT's $m, \sigma, \varepsilon/k$) goes in a `model`-keyed sub-block on the component, `eosParameters{ PCSAFT{...} }`, each factory reading *its own* sub-block and ignoring the rest.
3. PAIR parameters (k_{ij}) go in a model-keyed catalogue, `data/standards/eos/<model>/binaryInteractions/<i>-<j>.dat`, with the always-permitted inline override in the package.

A model with no required `eosParameters.<model>` block **fails with a remedy**, never a silent corresponding-states fallback.

25 The elements datum — one enthalpy surface

What you'll learn. Why the energy balance runs on ONE datum, how the heat of reaction / solution / crystallisation emerge from it, and what throws when a datum is missing.

A heat balance that has to close *through* a reactor, a crystalliser, or a dissolution cannot use a sensible (arbitrary-T_{ref}) datum — it would silently drop the heat of reaction. Choupo runs the energy ledger on the **elements datum**: formation enthalpies at 298.15 K, the same reference as the ideal-gas H_{ig}, so the reaction / phase-change / dissolution enthalpy *emerges* as the difference of formation enthalpies.

Key idea. One datum, no silent fallback (Vítor's law: elements-only, throw-on-gap). The second datum and every silent path were deleted — `streamH_sensible`, `solidH_sensible`, the `tryDatum(elements = false)` fallback, the `EnergyRef::Sensible` arm. `streamH_elements` now *propagates a real missing-Hf gap*, naming the component, instead of swallowing it. A curation gap is loud, never a quietly wrong closure.

The solid term. A stream's enthalpy is computed over both phases: the fluid (`s.F`, `s.z`) **and** the solid carried in `s.s[]` (kmol/s per component), on the same elements datum — $\sum_i s.s[i] \cdot h_i^{\circ}(\mathrm{solid}, T)$, the same solid formation leg `Component::h_formation(T,"solid")` uses. Before this, the crystals a crystalliser had just made were silently dropped from the stream enthalpy and the heat of crystallisation read zero; counting `s.s[]` (plus the aqueous solution tier for the dissolved solute) closed the crystalliser balance.

Utility legs vs process streams — no double-count. The ledger partitions each unit's ins/outs by `category` (`src/reporting/BalanceMath.H`, `EnergyBalanceReport.cpp`): utility-tagged legs (steam → condensate) feed a `qBoundary` accumulator — the real heat crossing the boundary — while process streams feed $h_{\mathrm{in}}/h_{\mathrm{out}}$. An evaporator's `duty_kW` is delivered *through its own steam stream*, so it is INTERNAL and must NOT also be summed as an external item; a process-to-process exchanger's duty is likewise internal. Closure $= 100 \cdot \Delta H/\mathrm{supplied}$ over genuine boundary heat only. This killed a fake 2131 % non-closure that was a pure double-count, not lost latent heat.

One enthalpy surface everywhere. The published stream `.H` (Flowsheet) and the isothermal-flash duty (`IsothermalFlash.cpp`) now run on the *same* canonical elements-datum form the report reads (`H_stream_formation`, $h_{ig} - \Delta h_{\mathrm{vap}}(T)$), not a Watson-at-298 + $\int c_p^{\mathrm{liq}}$ variant. The two differed by the liquid latent model and surfaced as a ~ 24 % gap between a flash's reported Q and its own stream ΔH; unifying the surface moves only the reported *duty* (an energy KPI), never the material flows.

26 Extending Choupo — step by step

> *What you'll learn.* Two complete, copy-pasteable walkthroughs — *tim-tim-por-tim-tim* — each ending in a green `bin/runTests`. Recipe A: a new unit operation, end to end. Recipe B: a new property-prediction method, with the validation-first step that earns your trust before you simulate with it.

26.1 Recipe A — a new unit operation, end to end

The worked example: `isothermalSplitter` — a unit that splits one feed into two outlets by a fraction θ to the top, both outlets at the feed's T, P, z (composition unchanged; the spec is the split fraction). Trivially simple physics on purpose — so every *mechanical* step is visible.

Step 1 — The header. `src/unitOperations/mixer/IsothermalSplitter.H` (it belongs beside the existing `Splitter`):

```
#pragma once
#include "core/Dictionary.H"
#include "thermo/ThermoPackage.H"
#include "unitOperations/UnitOperation.H"

namespace Choupo {

class IsothermalSplitter : public UnitOperation {
public:
    IsothermalSplitter() = default;
    const std::string& type() const override { return type_; }

    int solve(const DictPtr& dict,
              const ThermoPackage& thermo,
              int verbosity) override;

    std::vector<ProcessStream>      producedStreams() const override { return produced_; }
    std::map<std::string, scalar>   kpis()            const override { return kpis_; }

private:
    inline static const std::string type_ = "isothermalSplitter";
    std::vector<ProcessStream>      produced_;
    std::map<std::string, scalar>   kpis_;
};

} // namespace Choupo
```

Note the exact `solve` signature — `int solve(const DictPtr&, const ThermoPackage&, int)` — and that `type()` returns a reference to a member string. Prepend the licence banner (§on header banner); it doubles as the licence notice.

Step 2 — The implementation. `src/unitOperations/mixer/IsothermalSplitter.cpp`. The Flowsheet injects the inlet and your parameters as sub-dicts of `dict`; you publish outlets into `produced_` in the order your `outputs (...)` declares:

```
#include "mixer/IsothermalSplitter.H"
#include "core/Dimensions.H"
#include <iostream>

namespace Choupo {

int IsothermalSplitter::solve(const DictPtr& dict,
                              const ThermoPackage& thermo,
                              int verbosity)
{
    auto feed = dict->subDict("feed");          // F (kmol/s SI), Tfeed, Pfeed
```

```
    auto comp = dict->subDict("composition");   // mole fractions z_i
    auto op   = dict->subDict("operation");     // YOUR parameters

    const scalar F     = feed->lookupScalar("F",     Dims::molarFlow);
    const scalar Tfeed = feed->lookupScalar("Tfeed", Dims::temperature);
    const scalar Pfeed = feed->lookupScalar("Pfeed", Dims::pressure);
    const scalar theta = op->lookupScalar("splitFraction");    // [-], 0..1

    if (theta < 0.0 || theta > 1.0)
        throw std::runtime_error(
            "isothermalSplitter: splitFraction must be in [0,1]");

    std::vector<scalar> z(thermo.n());
    for (std::size_t i = 0; i < thermo.n(); ++i)
        z[i] = comp->lookupScalar(thermo.comp(i).name());

    if (verbosity >= 3)
        std::cout << "  [isothermalSplitter] F = " << (F * 3600.0)
                  << " kmol/h, split " << theta << " to top\n";

    produced_.clear();                          // ORDER == outputs(...) order
    ProcessStream top;  top.name = "top";
    top.F = theta * F;  top.T = Tfeed;  top.P = Pfeed;  top.z = z;
    produced_.push_back(top);

    ProcessStream bot;  bot.name = "bottom";
    bot.F = (1.0 - theta) * F;  bot.T = Tfeed;  bot.P = Pfeed;  bot.z = z;
    produced_.push_back(bot);

    kpis_.clear();                              // KPIs feed sizing/costing/sweep
    kpis_["F_top"]         = top.F;
    kpis_["F_bottom"]      = bot.F;
    kpis_["splitFraction"] = theta;

    return 0;                                   // 0 == converged
}

} // namespace Choupo
```

Key idea. **Fill EVERY field of a produced stream** (`F`, `T`, `P`, `z`, and `vf` / `s` if relevant). A half-filled stream propagates NaN downstream and the NaN guard fails the case — by design.

Step 3 — Register it (the one line). Edit `src/unitOperations/UnitOperation.cpp`: add the include beside the other mixer includes, and one line in `registerBuiltins()`:

```
#include "mixer/IsothermalSplitter.H"       // (1) beside #include "mixer/Splitter.H"
//...
void UnitOperation::registerBuiltins()
{
    auto reg = [](const std::string& name, auto&& maker) {
        registerType(name, std::forward<decltype(maker)>(maker)); };
    //...
    reg("splitter",           []{ return std::make_unique<Splitter>(); });
    reg("isothermalSplitter", []{ return std::make_unique<IsothermalSplitter>(); }); // (2)
    //...
}
```

That is the *entire* registration story — no macro, no static initialiser. A student greps `registerBuiltins` and sees your type listed.

Step 4 — Build (the recursive pickup).

```
make all
```

The Makefile discovers sources with `find src -name '*.cpp'`, so the new `.cpp` is compiled with no Makefile edit; `-MMD -MP` tracks the new header dependency automatically.

Step 5 — Rebuild the WASM (MANDATORY).

```
make wasm-gui          # = choupoSolve + choupoProps, the two the GUI uses
```

Common pitfall. Skip this and the GUI throws `UnitOperation::New: unknown type 'isothermalSplitter'` even though the native binary runs the type fine. The WASM solver is a separate Emscripten compile; a new built-in only reaches the browser after a WASM rebuild (`make wasm-gui`). The incremental build invalidates on source, flag and standards-content changes — a "Nothing to be done" after a real change is a graph bug, not something to paper over with ritual cleans (`make wasm-clean` stays available as deliberate diagnostics). Never run two `make wasm` concurrently — they clobber `gui/public/wasm/`.

Step 6 — A minimal tutorial case.

Keep it trivial so it doubles as a regression test. `tutorials/steady/userops/splitter`

```
splitter01_isothermal/
  splitter01.cho                # GUI marker (openable; may carry GUI layout)
  system/
    controlDict                 # application choupoSolve; verbosity 3;
    flowsheetDict               # the topology below
  constant/
    thermoPhysPropDict          # v2 record: components + gammaPhi ideal
```

The `system/flowsheetDict` (one feed boundary, one unit, two product boundaries):

```
boundary
{
    inlets  ( feed );
    feed { F 1.0 kmol/s; Tfeed 350 K; Pfeed 1 bar;
            composition { benzene 0.5; toluene 0.5; } }
}

units
(
    {
        name      S1;
        type      isothermalSplitter;
        in        feed;
        outputs   ( top bottom );        // ORDER matches produced_ order
        operation { splitFraction 0.3; }
    }
);
```

Step 7 — Add KPIs to the regression (`expected`).

Your `kpis()` already feed sizing, costing and the sweep/optimisation drivers; pin the load-bearing ones to golden masters. Drop a `tutorials/steady/userops/splitter01_isothermal`

```
# kind   name   key            expected   reltol
kpi      S1     F_top          0.3        1e-6
kpi      S1     F_bottom       0.7        1e-6
stream   top    F              0.3        1e-6
```

then record/verify and run the full gate:

```
bin/runTests --record tutorials/steady/userops/splitter01_isothermal  # write from a trusted run
bin/runTests --witnesses                      # rung 3: every execution class
```

Key idea. **The gate is `bin/runTests`, not a bare exit-code loop** — NaN/inf guard on *every* case plus the `expected` KPI/stream comparison (§16). A solver can return NaN with exit code 0; only the structured-result scan catches it.

26.2 Recipe B — a new property-prediction method

A property model plugs into one of four bases. **Pick the base by the quantity you predict**, then follow the same `fromDict` + `registerModel` pattern — and validate against measured data *before* you trust it.

You predict…	Base	`registerModel` factory signature
$P^{\mathrm{sat}}(T)$	`VaporPressureModel`	`(const DictPtr&)`
$c_p(T)$	`HeatCapacityModel`	`(const DictPtr&)`
$\gamma_i(T, x)$	`ActivityModel`	`(const DictPtr&, const std::vector<std::string>&)`
φ_i, Z, H^{res}	`EquationOfState`	`(const DictPtr&, const std::vector<Component>&)`

B.1 — A vapour-pressure model (the simplest base). Say you add a two-parameter `Clapeyron` P^{sat}.

1. `src/thermo/vaporPressure/Clapeyron.{H,cpp}`, inheriting `VaporPressureModel`, override the pure virtual `scalar Psat_Pa(scalar T_K) const`; a constructor `Clapeyron(const DictPtr&)` reads the parameters (mirror `Antoine.cpp`, whose ctor is `Antoine::Antoine(const DictPtr&)`).
2. Register — include + one line in `VaporPressureModel::registerBuiltins()`:

```
#include "Clapeyron.H"
//...
void VaporPressureModel::registerBuiltins()
{
    registerModel("Antoine",
        [](const DictPtr& d) -> std::unique_ptr<VaporPressureModel>
        { return std::make_unique<Antoine>(d); });
    registerModel("AmbroseWalton",
        [](const DictPtr& d) -> std::unique_ptr<VaporPressureModel>
        { return std::make_unique<AmbroseWalton>(d); });
    registerModel("Clapeyron",                                        // new
        [](const DictPtr& d) -> std::unique_ptr<VaporPressureModel>
        { return std::make_unique<Clapeyron>(d); });
}
```

3. Select it in a component's `vaporPressure{ model Clapeyron; ... }` block.

B.2 — Compare against measured data FIRST (the AAD step). A property model is **not trustworthy until it has been checked against measured data**. `choupoProps` gives you two different ways to do that, and confusing them is easy, so take them one at a time.

(i) The overlay comparison. Declare model scans plus an `experimental {}` block naming a measured dataset. The run prints one average absolute deviation (AAD) per model per property:

```
choupoProps tutorials/props/compare/compare_vle_etoh_water
# >>>  Comparison 'etoh_water_1atm' (abscissa x[ethanol], 12 measured pts):
#        txy_ideal  T_bubble : AAD 5.35404 K   (12/12 pts)
#        txy_nrtl   T_bubble : AAD 0.349277 K  (12/12 pts)
#        txy_unifac T_bubble : AAD 0.407352 K  (12/12 pts)
```

There is deliberately **no PASS here, and no tolerance**. The overlay reports a distance; what counts as close enough is a judgement about the system and the use, and the engine does not have it. What the suite does hold

is that the distance has not *moved*: since 2026-08-12 those AADs are pinned by `aad` rows in the case's `expected`. That matters — the same case's header claimed a UNIFAC AAD of ∼2.6 K for as long as the arm existed, against its own output's 0.407 K, and nothing could catch it while the number lived only in prose.

(ii) The bench verdict. `heatTransferBench` is the other shape: it carries a `aadTolerance` the case DECLARES, and it does return a verdict.

```
# heatTransferBench, and ONLY this family, prints a pass/fail line:
#   AAD = 1.8 %   AAD tolerance = 3.0 %   -> PASS
```

This paragraph used to show that second listing under a `propertyScan1D` command — one operation's transcript attributed to another, so a reader following it would run a scan and wait for a line that never comes.

Key idea. The number that earns trust is the AAD against measured data, not that the model compiles. But an AAD is a *comparison*, not a validation: it says the model agrees with these data, never that the data were independent of the fit that produced its parameters. A model fitted to a series sits close to that series by construction — Choupo's own corpus has both kinds side by side, and `docs/architecture/verification-and-validation.md` labels each one. A verdict needs a band somebody DECLARED; where no band is declared, read the distance and decide yourself. That is the property half of "see, then decide": you simulate with a model *because* you have seen its AAD, never because a dropdown recommended it.

B.3 — An equation of state (and the data-doctrine path).

An EoS is the one base where WHERE the parameters live is a doctrine decision. The canonical reference is `docs/ai/data-doctrine.md` §4; in short:

1. **Corresponding-states EoS (reads only the triad).** If the model needs only $\{T_c, P_c, \omega\}$, **read them off the `Component` and derive the rest in source** — touch *zero* data files. This is exactly what `SRK` and `PR` do: the `SRK` constructor reads `c.Tc()`, `c.Pc()`, `c.omega()` and computes $a_c = \Omega_a R^2 T_c^2 / P_c$, $b = \Omega_b R T_c / P_c$, $m = m(\omega)$. Do *not* pre-bake a_c into the `.dat` (it would duplicate a value the model owns and let the two drift).
2. **Model-specific PURE parameters (cannot be generated).** PC-SAFT's $\{m, \sigma, \varepsilon/k\}$ live in a `model`-keyed sub-block on the component — additive, never disturbing the triad:

   ```
   // inside the component .dat
   eosParameters
   {
       PCSAFT  { m 2.4653;  sigma 3.6478e-10;  epsilon_k 287.35;
                 provenance { origin regressed; method "Gross & Sadowski 2001"; } }
       // SRK/PR appear in NO key here -- they read Tc, Pc, omega
   }
   ```

   Each factory reads *its own* `eosParameters.<model>` sub-block by name and ignores the rest, so one molecule carries SRK + PR + PC-SAFT simultaneously, each with its own `provenance{}`.
3. **Pair parameters k_{ij}.** A model-keyed catalogue, `data/standards/eos/<model>/binaryInteractions/<i>-<j>.dat`, with the always-permitted inline override. `SRK::fromDict` already reads an inline `binaryInteractions` list of `{ i; j; kij; }` dicts and resolves names positionally against the component list.

The explicit factory, and the open follow-up.

Register the EoS the same explicit way, dispatching to a `fromDict` static (the `Component`-list overload):

```
registerModel("PCSAFT",
    [](const DictPtr& dict, const std::vector<Component>& comps)
        -> std::unique_ptr<EquationOfState>
    { return PCSAFT::fromDict(dict, comps); });
```

Common pitfall. **Glass-box requirement:** an EoS with no required `eosParameters.<model>` block on a component must **fail with a remedy** ("PCSAFT needs m, sigma, epsilon_k for X; fit them with choupoProps or pick SRK/PR which run off Tc, Pc, omega"), *never* a silent corresponding-states fallback. **Open follow-up** (noted in the doctrine): a uniform `modelParams` accessor on `Component` that hands a factory

its `eosParameters.<model>` sub-dict by model name is not yet generalised — today each EoS that needs a pure block reads it itself. Until that lands, copy the `liquidViscosity{}` raw-sub-dict reader pattern.

27 Roadmap

Ordered by impact for teaching and for Vítor's research:

Target	Idea
(shipped)	Levenberg-Marquardt parameter estimation (`fitParameters`, in `choupoProps`; the `fitBinaryPair` outer dri Nelder-Mead optimisation driver (minimise cost / TAC) LL flash via 2nd-order Michelsen TPD with multiple restarts Solids: inert phase + filter / cyclone / dryer unit-ops Crystalliser + PSD basics NRTL distillation robustness (Naphtali-Sandholm)
v1.0	Equation-oriented sparse Newton for the full flowsheet
> v1	Membrane NF/RO unit op (Vítor's research bridge)

Tech-debt items also welcome:

- Reference logs for every tutorial → in-tree regression diff.
- Reaction-kinetics library (Arrhenius, Langmuir-Hinshelwood, Michaelis-Menten) as a `Kinetics` base class.

28 Things to NOT do

> *What you'll learn.* A short list, each item born from a real incident. Memorise it.

- Do **not** suggest a Python rewrite or wrapper. The course teaches numerical methods in C++; rewriting in Python would defeat the purpose.
- Do **not** add macro auto-registration. Explicit `registerBuiltins()` is non-negotiable (see §6).
- Do **not** add a heavy dependency — even a "small" one. Pedagogical visibility outweighs convenience.
- Do **not** split a binary by numerical strategy (`*Foam`-style proliferation). Strategy is run-time. Splitting by *problem class* — the five binaries Solve / Batch / SemiContinuous / Ctrl / Props — is the deliberate exception, not a precedent for more.
- Do **not** bypass the dict format. Never use YAML, JSON, or TOML inside the case directory. The dict is the canonical wire format.
- Do **not** relitigate decisions in `CLAUDE.md` §10 without first reading what's already been discussed and why.
- Do **not** merge code that breaks the regression, even if the failure is "obviously unrelated".
- Do **not** add a directory under `src/` without declaring its band (§19). A subsystem with no band fails the layering gate on purpose — defaulting it to the bottom would put it where nothing can violate.
- Do **not** store a derived quantity beside its inputs (§20). Compute it; keep any declared value as an anchor the run compares aloud.
- Do **not** put a behaviour-changing fact in a comment or a banner. If the engine cannot parse it, it does not exist.
- Do **not** ship a gate you have not sabotaged (§21). A gate that survives the break it names is making a claim it cannot support.

29 Quick reference

Task	Command / file
Build release	`make all`
Build debug	`make MODE=debug all`
Clean	`make clean` / `make distclean`
Source env	`source etc/bashrc`
Run a case	`runCase tutorials/<name>`
Run from inside case dir	`runCase`
List tutorials	`listCases`
List built-in unit-op types (in code)	`UnitOperation::availableTypes()`
Tweak a dict field programmatically	`dict->setScalarAtPath(path, v)`
Add a unit op (full walkthrough)	§26.1 (quick: §7)
Add a property model (full walkthrough)	§26.2 (quick: §9)
Add a component / material / reaction	§12
Add an outer driver / post-processor	— (see the outer-driver section)
Where does a number live?	§24 + `docs/ai/data-doctrine.md`
Which subsystem may include which?	§19 + `docs/architecture/module-boundaries.md`
How a class is shaped (the five rules)	§20 + `docs/architecture/class-structure.md`
Writing and sabotaging a gate	§21
Where an accepted violation is recorded	`bin/curate/debt_registry.py`
Energy datum / balance closure	§25
Sectored flowsheet + instants	§22, §23
Project conventions / AI rules	`CLAUDE.md`
User-facing how-to	`docs/userGuide.pdf`

Colophon. This manual was typeset in Latin Modern with the Choupo shared LaTeX preamble. Sources live in `docs/`; rebuild with `cd docs && make`.